CNN-Based Hyperspectral Pansharpening With Arbitrary Resolution

Lin He, *Member, IEEE*, Jiawei Zhu, Jun Li, *Fellow, IEEE*, Antonio Plaza, *Fellow, IEEE*, Jocelyn Chanussot, *Fellow, IEEE*, and Zhuliang Yu, *Member, IEEE*

***Abstract*— Traditional hyperspectral (HS) pansharpening aims at fusing a HS image with its panchromatic (PAN) counterpart, to bring the spatial resolution of the HS image to that of the PAN image. However, in many practical applications, arbitrary resolution HS (ARHS) pansharpening is required, where the HS and PAN images need to be integrated to generate a pansharpened HS image with arbitrary resolution (usually higher than that of the PAN image). Such an innovative task brings forth new challenges for the pansharpening technique, mainly including how to reconstruct HS images beyond the training scale and how to guarantee spectral fidelity at any spatial resolutions. To tackle the challenges, we present a novel convolutional neural network (CNN)-based method for ARHS pansharpening called ARHS-CNN. It is based on a two-step relay optimization process, which is associated with a multilevel enhancement subnetwork and a rescaling subnetwork. With a careful design following the thread, our ARHS-CNN is able to pansharpen HS images to any spatial resolutions using just a single CNN model trained on a limited number of scales while meantime to keep spectral fidelity at those resolutions, which wins an obvious advantage over traditional pansharpening methods. Experimental results obtained on several datasets verify the excellent performance of our ARHS-CNN method.**

***Index Terms*— Arbitrary resolution HS (ARHS) pansharpening, convolutional neural networks (CNNs), dynamic convolution, hyperspectral (HS) pansharpening, spectral fidelity.**

NOMENCLATURE

$\widetilde{\mathbf{H}}$	Low-resolution HS (LRHS) image.
$\mathbf{P}$	Panchromatic (PAN) image.
$\widetilde{\mathbf{H}}^{\uparrow}$	Pre-interpolated LRHS image.
$\mathbf{H}_{\mathrm{P}}$	Standard HS pansharpening reference image.
$\widehat{\mathbf{H}}_{\mathrm{AR}}$	Reconstructed arbitrary-resolution HS (ARHS) image.
$\mathbf{H}_{\mathrm{AR}}$	Ideal ARHS image.
$\mathbf{H}_{\mathrm{TR}}$	Training high-resolution HS data.
$\widehat{\mathbf{H}}_{\mathrm{TR}}$	Reconstructed training high-resolution HS data.
s	ARHS pansharpening scale, i.e., the spatial resolution ratio between HS images before and after ARHS pansharpening.
r	Spatial resolution ratio between the PAN and LRHS images.
s'	Intermediate scale, i.e., the spatial resolution ratio between the ARHS pansharpened image and the PAN image.

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I. INTRODUCTION

IMAGING spectrometers can form images with many spectral bands. The resulting hyperspectral (HS) images possess the ability to pixel wisely accommodate spectral behaviors of observations [1]–[3], which is of great value in a multitude of remote sensing situations, including classification [3], unmixing [4], target/change detection [5], [6], etc. However, there are crucial compromises among their spatial and spectral resolutions, and other factors such as signal-to-noise ratio in such remote-sensing systems. Usually, they can yield data either with high spatial resolution but low spectral resolution (e.g., PAN images) or with a large number of spectral bands but low spatial resolution (typical examples include HS images [2]). Either PAN images or HS images maybe insufficient to deal with practical applications which require images with high spatial and spectral resolutions. One solution to this problem is to employ HS pansharpening technique, which aims at integrating low (spatial) resolution HS (LRHS) images with the connected high (spatial) resolution PAN images to yield high (spatial) resolution HS (HRHS) images.

In its early stage, pansharpening mainly refers to enhancing the spatial resolution of multispectral (MS) images by integrating PAN images, which have been extended to inspire HS pansharpening methods [2], [7]. Representative pansharpening approaches can be largely divided into several groups. The component substitution (CS) category operates a replacement of an HS image's component with the PAN

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image, usually after a domain conversion [8]–[10]. The multiresolution analysis (MRA) group often involves detail injection in a multiresolution manner [11]–[15]. A hybrid method called guided filter-based principal component analysis was proposed in [16] to combine the benefits of both CS and MRA. Model-based methods consider pansharpening as a problems related to observation models and ill-posed inverse optimization problems, which can be solved by involving Bayesian estimation [17]–[20], sparse representation [21]–[23], coupled nonnegative matrix factorization (CNMF) [24], primal-dual splitting [25], and the alternating direction multipliers [26], [27].

Convolutional neural network (CNN)-based pansharpening is presently under intensive study [28]–[36], which is a notable instance of the learning-based methodology. Although involving a training (learning) process and thus usually being relatively complicated to run, learning-based approaches need less model assumptions and offer the potential to mine useful patterns and structures in the complex data. More specifically, CNN pansharpening usually trains a CNN model at the reduced resolution and then applies it to the full resolution to yield pansharpened HS images, which is under the hypothesis of scale invariance [28], [33], [37]. Masi *et al.* [28] pioneered an architecture that stack the pre-interpolated MS images with their associated PAN images as the inputs of the lightweight CNN, further improving the network performance utilizing a target-adaptive modality [31]. Wei *et al.* [29] and Yuan *et al.* [30] introduced a multi-scale kernel trick to the residual CNN structure for pansharpening. He *et al.* [32], [33], and [38] designed pansharpening CNNs (PNNs)-based methods involving the considerations of detail injection, spectral preservation, and spatial feature transform.

As far as HS pansharpening is concerned, some intriguing problems need to be addressed, including the significant spectral resolution distinction between the HS and the PAN data, and that spatial details have to be synthesized in many continuous narrow spectral bands. Some CNN-based approaches have been proposed recently for HS pansharpening. Li *et al.* employed a deep Laplacian-related PNN to pre-reconstruct LRHS images in a band-by-band manner and then added details on the images. Xie *et al.* [36] used CNN to learn priors of a pansharpening relevant constrained minimization. He *et al.* [35] developed a constrained adversarial network to extract deep features for further pansharpening. In an attempt to alleviate notorious spectral distortions, He *et al.* [33] developed spectral predictive CNN structures achieving excellent performance.

Over recent years, the usage of pansharpening for remote-sensing applications has been explored, such as enhancing thematic mapping [39]–[41], change detection [42], [43], object detection [44], [45], land use/cover classification [41], [46], agriculture management [47], [48], and visual interpretation [2], [49]. In these applications, pansharpening, which seeks to adjust the data resolution, in fact serves for the final tasks as an intermediate process. At this point, it is noteworthy that almost all existing pansharpening methods are aimed at enhancing the spatial resolution of a MS/HS image to match the resolution of the corresponding PAN image. However, in aforementioned practical applications, users may need a product with different spatial resolution from the available MS/HS image and the connected PAN image.

Take thematic mapping for example, which may be one of the most representative applications that pansharpening can work for. Spectral image-based thematic mapping has been systematically studied in the literature [39], [40], [46], [50]–[53], where special-purpose information extracted by involving spectral images can be graphically illustrated in the form of images conveying the spatial variability of themes. Obviously, spatial resolution is one of necessary elements of a thematic map, which implies a certain level of information. Therefore, pansharpening technique can be utilized to produce higher resolution or higher level thematic maps by enhancing the spatial resolutions of spectral images [40], [46]. When we utilize standard HS pansharpening to enhance thematic mapping, the resulting thematic maps are with the spatial resolution predetermined by the PAN. Such a manner of thematic mapping is obviously limited and not adaptive, considering a wide range of potential thematic map applications which may involve users with different spatial resolution requirements. For instance, users may require a thematic map with the spatial resolution of 0.1, 0.25, or 0.52 m while, say, only 10-m resolution HS images and 2.5-m resolution PAN images are available. In this situation, ARHS panshapening offers a feasible way to support the subsequent tasks, i.e., integrating the HS and the PAN images to yield an HS image with the desire resolution (may be different from the PAN resolution) and then accordingly producing a thematic map with this resolution. In addition, thematic mapping validation may be another instance that ARHS pansharpening could be useful for thematic mapping [54]. As a kind of remote-sensing products, spectral-based thematic maps may need a validation processing before putting into practical applications, which is a procedure to qualify their accuracies [8,9]. A possible way to perform the validation is to test the thematic maps if they are consistency with another group of thematic maps produced with spectral images captured by other sensors or platform while covering the same area. However, different imaging sensors and imaging platform may yield data with different spatial resolution. Our ARHS pansharpening method could freely tune the spectral data to arbitrary desired resolution and accordingly support producing thematic maps with any desired resolution, therefore ensure the fulfillment of the thematic map validation.

ARHS panshapening, as a new type of HS pansharpening task introduced in this work, faces some special challenges which are not presented in traditional HS pansherpening. For instance, how to reconstruct the HS image at any spatial resolutions that are beyond the training scale, or how to keep the spectral fidelity at any spatial resolution (where details in the PAN image cannot be directly injected to the LRHS image). In this article, we introduce an innovative CNN-based method for the new ARHS pansharpening task, called ARHS-CNN. To the best of our knowledge, there is no previous work in the literature addressing the issue of arbitrary resolution pansharpening. The major contributions of our work can be summarized as follows.

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1) A new CNN architecture able to conduct the new arbitrary-resolution pansharpening task. We develop a novel CNN architecture (and the connected implementation approach) able to pansharpen HS images to an arbitrary spatial resolution (which does not need to be predefined during the training phase). For this CNN, there is no demand for the consistency between the pansharpening scales connected with the training phase and with the pansharpening phase. Once an PNN model is trained in the training phase (with the degenerated data), it can be directly generalized to the succeeding pansharpening or testing phase to produce HS images with arbitrary resolutions for given observed HS images. That is, our PNN offers the potential to produce any user-desired spatial resolutions using just a single-trained CNN model while achieving remarkable spectral fidelity.
2) A rescaling subnetwork. Spatially dynamic convolution and spectral augmentation are incorporated to achieve spectral–spatial reconstructions at any arbitrary resolution levels. Especially, during the convolution operation, spatially dynamic kernels (inferred from a spatial offset-kernel mapping) are jointly utilized with band-wise dynamic convolution to tune the spatial resolution of the data, where the limitations imposed by the training scale can be overcome thanks to the spatial offset-kernel mapping mechanism.
3) A multilevel enhancement subnetwork. To fully accommodate the spatial structures of the PAN image, the LRHS and PAN images are first merged using a new initial fusion (IF) module whose output is further strengthened by the gradient maps of the PAN image in a multilevel manner. The multilevel enhancement subnetwork is then connected to the aforementioned rescaling subnetwork.
4) A composite loss. The loss is constructed by combining a reconstruction constraint, a multistage constraint, and a spatial constraint, which is beneficial to make the bottom features more interpretable, improving the reconstruction ability of the generated HS image with arbitrary resolution.

In summary, our newly developed ARHS-CNN method can accomplish pansharpening HS data to any spatial resolutions under a single CNN model, while keeping spectral fidelity at those resolutions. The rest of the article is organized as follows. Section II presents a general description of the problem and summarizes the important new challenges involved in ARHS pansharpening. A detailed presentation of our newly developed ARHS-CNN method is given in Section III. Our experimental results are presented and analyzed in Section IV. Section V concludes the article with some remarks.

II. Problem Description

ARHS pansharpening aims to enhance the resolution of an LRHS image to a user-customized resolution by fusing it with the connected PAN image, rather than merely bringing the spatial resolution of the HS data to that of the PAN image (as traditional HS pansharpening methods do). Such a pansharpening task is able to transfer LRHS images to any resolutions, providing a very flexible framework that exhibits the potential to be useful in many practical applications.

As a new type of pansharpening, besides the problems of the striking difference between the spectral resolution of the HS and the PAN images and having to recover details in many continuous bands, ARHS pansharpening has to deal with other specific challenges described as follows.

1) How to reconstruct HS images beyond the training scale? Hereinafter, the resolution ratio between HS images before and after ARHS pansharpening is called ARHS pansharpening scale. Many state-of-the-art pansharpening methods, especially CNN-based ones, depend on a two-stage process to accomplish pansharpening. Given the LRHS data and the associated PAN data, they first train a resolution-enhancement model in a scale, and then fulfill the final pansharpening with this model. However, in ARHS pansharpening (where the target spatial resolution of the pansharpened HS image is user-customized and not predefined), there may be many possible desired resolutions and, thus, desired ARHS pansharpening scales which could be integer or non-integer. In this situation, it is not convenient (or even impossible) to train a specific model for each candidate scale, due to the fact that this would result in expensive computations and unaffordable storage requirements. Therefore, it is required for a practical ARHS pansharpening method to be able to reconstruct HS images to any scales (i.e., pansharpen LRMS images to any spatial resolutions) under a single model obtained using training data in one or a very limited number of scales.
2) How to guarantee spectral fidelity at any spatial resolutions, including those that are much higher than that of the PAN image? Traditional HS pansharpening is devoted to improving the spatial resolution of an LRHS image to that of the connected PAN image, while ARHS pansharpening aims at zooming the LRHS image to any resolution. In the latter scenario, the PAN image may have a resolution that is different from the target resolution. In this case, the PAN image tends to provide less support to the task of recovering the HS image at that particular target resolution, being prone to introduce spectral distortions (especially when the target resolution is much higher than the PAN resolution).

III. Proposed Method

In this section, we propose a CNN-based method for the new ARHS pansharpening task, called ARHS-CNN. It intends to solve the two main problems specifically associated with ARHS pansharpening, described in above Section II.

Let $\mathbf{H} \in \mathbb{R}^{h\times w\times b}$ be an LRHS image and $\mathbf{P} \in \mathbb{R}^{hr\times wr}$ be the connected PAN image, where $h \times w$ represents the spatial size, b denotes the number of spectral channels, and r is the spatial resolution ratio between the PAN and LRHS images. ARHS pansharpening aims to achieve an ARHS image $\mathbf{H}_{\mathrm{AR}} \in \mathbb{R}^{sh\times sw\times b}$ (i.e., $\in \mathbb{R}^{rs'h\times rs'w\times b}$), where s is the ARHS pansharpening scale and $s' = \frac{s}{r}$ is an intermediate scale

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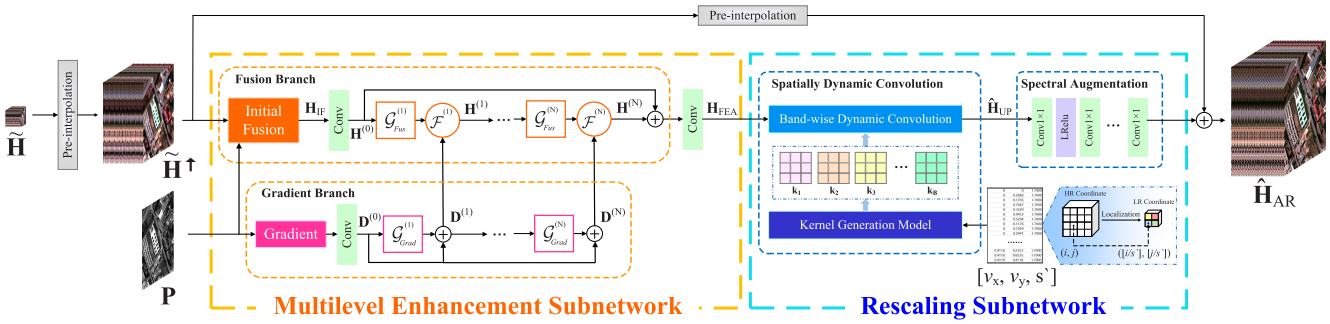


Fig. 1. Overall architecture of the proposed ARHS-CNN. ⊕ denotes element-wise sum.

that is the spatial resolution ratio between the ARHS and PAN images. s can be any value determined by the practical arbitrary resolution requirement, either integer or non-integer, usually being greater than r. When s equals r, ARHS pansharpening is degenerated to traditional HS pansharpening.

We formulate ARHS pansharpening as a two-step relay optimization problem

$$\arg\min_{\boldsymbol{\theta}_1,\boldsymbol{\theta}_2} \mathcal{L}\{\mathcal{U}_{\boldsymbol{\theta}_2}(\mathcal{M}_{\boldsymbol{\theta}_1}(\widetilde{\mathbf{H}},\mathbf{P}),s'),\mathbf{H}_{\mathrm{AR}}\} \tag{1}$$

where $\mathcal{M}_{\boldsymbol{\theta}_1}(\cdot)$—associated with the first step—denotes the mapping from the LRHS and PAN images to the intermediate data with the same spatial resolution as the PAN image, $\mathcal{U}_{\boldsymbol{\theta}_2}(\cdot)$—connected to the second step—is an operation to yield the final ARHS image based on $\mathcal{M}_{\boldsymbol{\theta}_1}(\cdot)$. In $\mathcal{M}_{\boldsymbol{\theta}_1}(\cdot)$, the LRHS image and the PAN image are jointly involved in the generation of an intermediate data with the spatial resolution of the PAN image, which is beneficial to fully and directly utilize spatial structures of the PAN image, and to the subsequent reconstruction of the ARHS image. In $\mathcal{U}_{\boldsymbol{\theta}_2}(\cdot)$, the intermediate data yielded with $\mathcal{M}_{\boldsymbol{\theta}_1}(\cdot)$ is zoomed (rescaled) along the spatial and spectral dimensions, to achieve the final ARHS image. For convenience, we describe the related notations in Nomenclature.

A. Overall Architecture

Our ARHS-CNN is mainly composed of two subnetworks, which is in line with the two-step relay process formulated in (1). Its overall architecture is graphically illustrated in Fig. 1. As can be seen in the figure, the two subnetworks are: 1) the multilevel enhancement subnetwork and 2) the rescaling subnetwork, which are related to $\mathcal{M}_{\boldsymbol{\theta}_1}(\cdot)$ and $\mathcal{U}_{\boldsymbol{\theta}_2}(\cdot)$ in (1), respectively. They are embedded in a unified CNN framework, so that their network parameters can be updated in an end-to-end manner when ARHS-CNN is running.

B. Multilevel Enhancement Subnetwork

This subnetwork is dedicated to enhancing the resolution of the LRHS image to the level of the PAN image, which is advantageous to fully and directly utilizing spatial structures of the PAN image and, thus, to the final arbitrary resolution reconstruction. It consists of a fusion branch and a gradient branch, which are incorporated to merge spatial details from the PAN image in a multilevel way.

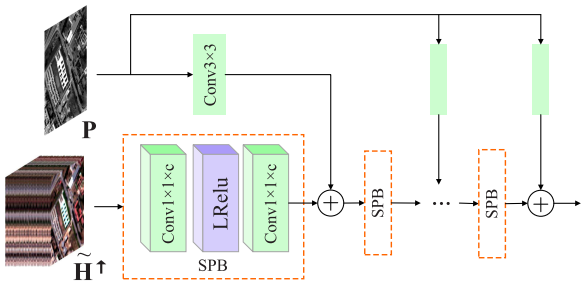


Fig. 2. Diagram of the IF module.

1) Fusion Branch: The LRHS image $\widetilde{\mathbf{H}}$ is first pre-interpolated to the same spatial size as the PAN image $\mathbf{P}$ [28], [29], [33]. Then, the interpolated HS image $\widetilde{\mathbf{H}}^{\uparrow}$ and its panchromatic counterpart are concurrently passed to an IF module, whose structural details are given in Fig. 2. It is well known that neighboring bands of HS images are highly correlated in the spectral domain [55], and thus it is reasonable to generate representative bands within a given band distance c. Here, we designed a spectral preservation block (SPB) based on 3-D convolution to effectively reduce redundant bands while preserving spectral fidelity. In the meantime, spatial features formed from $\mathbf{P}$ are added to SPB outputs. Due to the lightweight structure of SPB, it is feasible to stack multiple SPBs in the IF module. As the IF module is fully convolutional and parameter-differential while able to work with any number of input bands, it exhibits the potential to be flexibly embedded into any HS PNNs.

The IF module roughly fuses spectral and spatial features at shallow levels in our CNN. Its output $\mathbf{H}_{\mathrm{IF}}$ is subsequently processed with a residual-nested structure, where we use residual-in-residual dense blocks (RRDBs) [56] in the main path. Such a structure implies performing residual learning in multiple levels which is beneficial to feature extraction. As shown in Fig. 1, each of RRDB group $\mathcal{G}_{\mathrm{Fus}}^{(n)}$ and RRDB group $\mathcal{G}_{\mathrm{Grad}}^{(n)}$, which are in the fusion branch and in the gradient branch, respectively, could comprise several stacked RRDBs. The group $\mathcal{G}_{\mathrm{Fus}}^{(n)}$ and the group $\mathcal{G}_{\mathrm{Grad}}^{(n)}$ are used in pairs, and the output of $\mathcal{G}_{\mathrm{Fus}}^{(n)}$ is integrated with the gradient branch (as shown in Fig. 1) to conduct multilevel fusion.

2) Gradient Branch: To reinforce the spatial structures from the PAN image during the fusion, we introduce a gradient

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branch into this subnetwork. As shown in Fig. 1, the gradient branch takes the PAN image $\mathbf{P}$ as the input and outputs its deep gradient features, wherein a shallow gradient map is first obtained with the differential operation

$$\mathbf{D} = \|(D_x, D_y)\|_2 \tag{2}$$

and then passed into the further processing, where $D_x = \mathbf{P}(x+1, y) - \mathbf{P}(x-1, y)$, $D_y = \mathbf{P}(x, y+1) - \mathbf{P}(x, y-1)$ and $\|\cdot\|_2$ denotes pixel-wise 2-norm. In this way, the branch tends to concentrate more on the spatial variation and thus more on representative spatial details in the observed scene. Then, layer-wise features from the gradient branch collaborate with those from the fusion branch in the following manner:

$$\begin{aligned}\mathbf{H}^{(n)} &= \mathcal{F}^{(n)}\left(\mathcal{G}_{\text{Fus}}^{(n)}(\mathbf{H}^{(n-1)}), \mathbf{D}^{(n)}\right)\\ \mathbf{D}^{(n)} &= \mathcal{G}_{\text{Grad}}^{(n)}(\mathbf{D}^{(n-1)}) + \mathbf{D}^{(0)}\end{aligned} \tag{3}$$

where $\mathcal{G}_{\text{Grad}}^{(n)}$ denotes the nth group of RRDBs in the gradient branch and $\mathcal{F}^{(n)}(\cdot)$ denotes concatenation and one-layer convolution, while $\mathbf{H}^{(n)}$ and $\mathbf{D}^{(n)}$ are variables as shown in Fig. 1. Since the gradient map carries rich spatial details, the gradient branch still holds the ability to compensate the loss of spatial structure in the fusion branch when the two branches deepen. Finally, the fusion branch and the gradient branch cooperate to yield the output of the multilevel enhancement subnetwork, i.e., spatial–spectral features $\mathbf{H}_{\text{FEA}}$ with the same spatial resolution as the PAN image (as presented in Fig. 1), which are the intermediate features that we will feed into the subsequent processing stage.

With such a design, our multilevel enhancement subnetwork is able to effectively integrate spectral information and spatial information in a multilevel fashion, enhancing the spectral–spatial features level-by-level in the PAN spatial resolution. It relieves the burden of integrating those two kinds of complex information in merely a single level of data while tends to fully utilize spatial structures of the PAN image and avoid spectral interference during the information integrating, thus achieving a fine and thorough fusion and tending to yield more representative spectral–spatial features used for the succeeding rescaling subnetwork.

C. Rescaling Subnetwork

The rescaling subnetwork is dedicated to generating the final ARHS image, which exhibits user-customized spatial resolution while the same number of spectral bands as the LRHS image. This is done by rescaling the intermediate spectral–spatial features $\mathbf{H}_{\text{FEA}}$ along both the spectral and spatial dimensions.

In particular, this subnetwork is able to enhance the resolution of the input data to any spatial resolution under a single training model. The subnetwork comprises two major operations: spatially dynamic convolution and spectral augmentation, as graphically illustrated in Fig. 3.

Further, to help better understand the practical implementation of our ARHS-CNN and the principle and the mechanism that underlie this method, especially how the rescaling subnework works, the connected data flow diagrams for both training phase and testing phase of ARHS-CNN are given as Fig. 4.

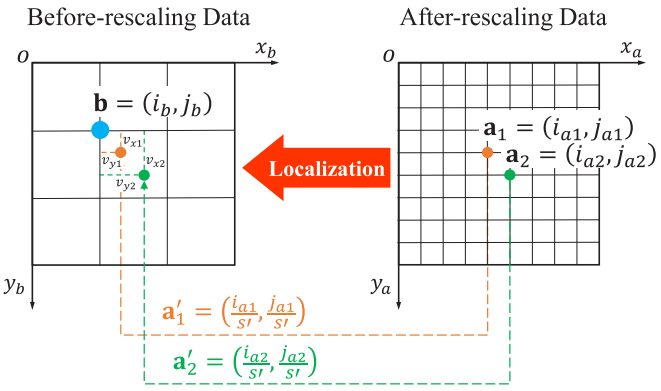


Fig. 3. Graphical illustration of spatially rescaling subnetwork. $\otimes$ denotes band-wise dynamic convolution.

1) Spatially Dynamic Convolution: The convolution operation is one of the most effective ways to rescale images, and has triggered enormous interest in the computer vision and image-processing community [57]–[59]. Here, we employ a spatially dynamic convolution to adjust the spatial resolution of the input data. The convolution approach comprises two basic steps: kernel generation and band-wise dynamic convolution, as shown in Fig. 3, where spatial kernels are first produced and then band-wise dynamic convolution is perform to yield relatively coarse arbitrary resolution data which is then fed to a subsequent spectral augmentation process. In this task, a offset-kernel mapping model is learned and then can be generalized to yield arbitrary resolution data.

a) Kernel generation: Fig. 5 graphically exemplifies the spatial relationship of pixels on the data before and after rescaling. As observed, two marked pixels of the after-rescaling data (i.e., the pixels $\mathbf{a}_1 = (i_{a1}, j_{a1})$ and $\mathbf{a}_2 = (i_{a2}, j_{a2})$ on the right figure) can be localized on the same pixel of the before-scaling data (i.e., the blue marked pixel $\mathbf{b} = (i_b, j_b)$ on the left figure) while they are connected to different spatial offsets $\mathbf{v}_1 = (v_{x1}, v_{y1})$ and $\mathbf{v}_2 = (v_{x2}, v_{y2})$. The spatial offset is computed as

$$\begin{aligned}\mathbf{v} &= (v_x, v_y)\\ &= \left(\frac{i_a}{s'} - i_b, \frac{j_a}{s'} - j_b\right)\\ &= \left(\frac{i_a}{s'} - \left\lfloor\frac{i_a}{s'}\right\rfloor, \frac{j_a}{s'} - \left\lfloor\frac{j_a}{s'}\right\rfloor\right)\end{aligned} \tag{4}$$

where v_x and v_y represent the x-direction and y-direction offsets, respectively, (i_a, j_a) denotes the spatial location of a pixel relative to the after-rescaling data with the intermediate scale s', (i_b, j_b) represents the location of that pixel on the before-rescaling data, and $\lfloor\cdot\rfloor$ is the floor function (giving the greatest integer value that less than or equal to the input). In other words, the offset $\mathbf{v}$ indicates the subpixel-level before-rescaling location of a pixel of the after-rescaling data. Therefore, it is implied that, if we use the spatial neighborhood of the pixel $\mathbf{b}$ of the before-rescaling data to form convolutions to constitute those two pixels $\mathbf{a}_1$ and $\mathbf{a}_2$ of the after-rescaling

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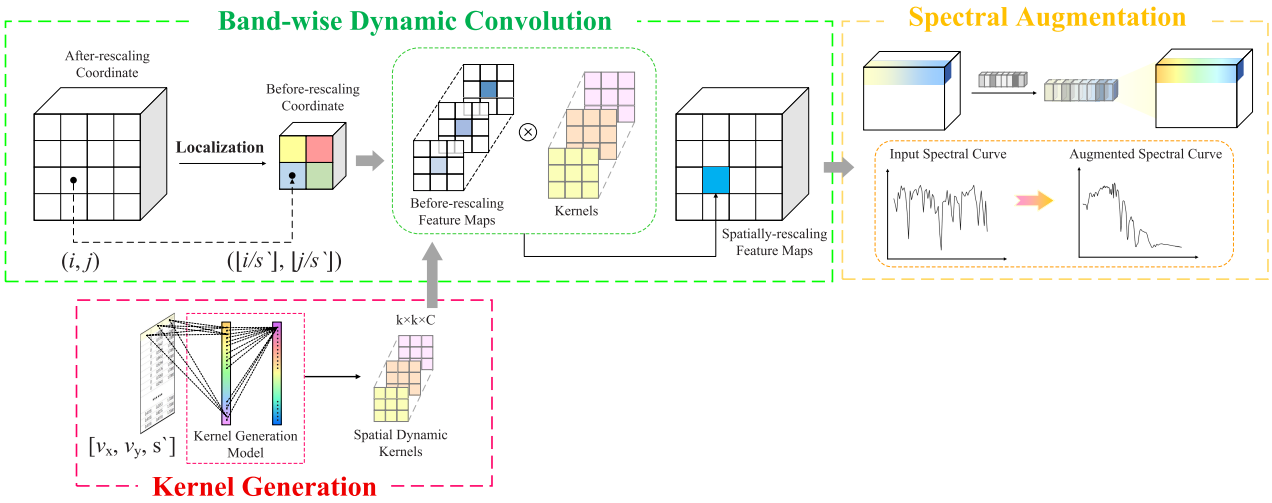


Fig. 4. Data flow diagram for training and testing phases of ARHS-CNN.

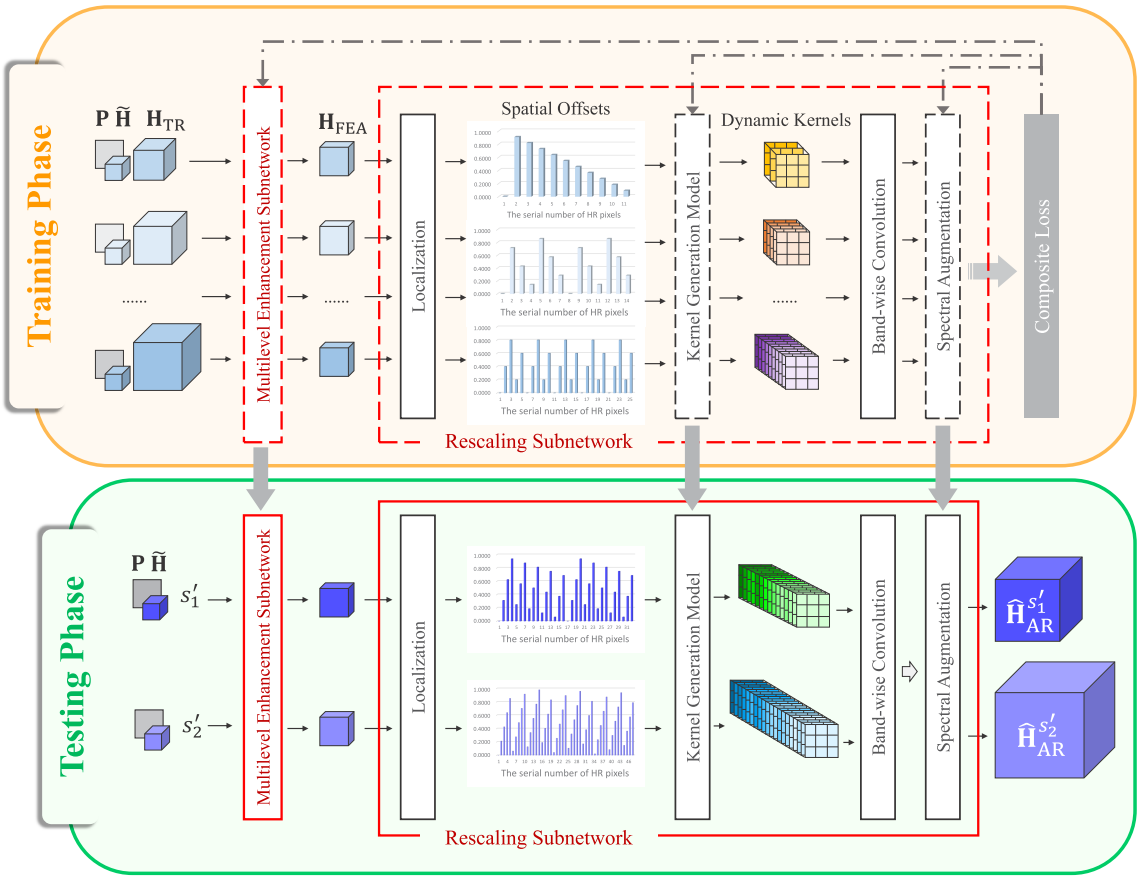


Fig. 5. Spatial arrangement of pixels before and after rescaling.

data, their connected kernels (i.e., the two sets of weights) should be different, since they are on different subpixel-level locations relative to the center pixel **b**. With this in mind, a relationship model mapping from spatial offset to spatial dynamic kernel can be formulated as follows:

$$\mathbf{k} = f_o(\mathbf{v}; \mathbf{W}) \tag{5}$$

where **k** represents the dynamic kernel used for the convolution to yield a pixel on the after-scaling data, $f_o(\cdot)$ stands for a local neural network building connection from the spatial offset **v** to **k**, **W** denotes the parameters of that local neural network and therefore $f_o(\cdot; \mathbf{W})$ represents the offset-kenel mapping (i.e., the kernel generation model as shown in Fig. 4). Once the parameters **W** have been optimized as $\mathbf{W}_0$ by a training process (in the training phase as presented in Fig. 4), the mapping from a spatial offset to its corresponding dynamic kernel is determined as $f_o(\cdot; \mathbf{W}_0)$. Whenever we need to rescale data to an arbitrary resolution (in the testing phase

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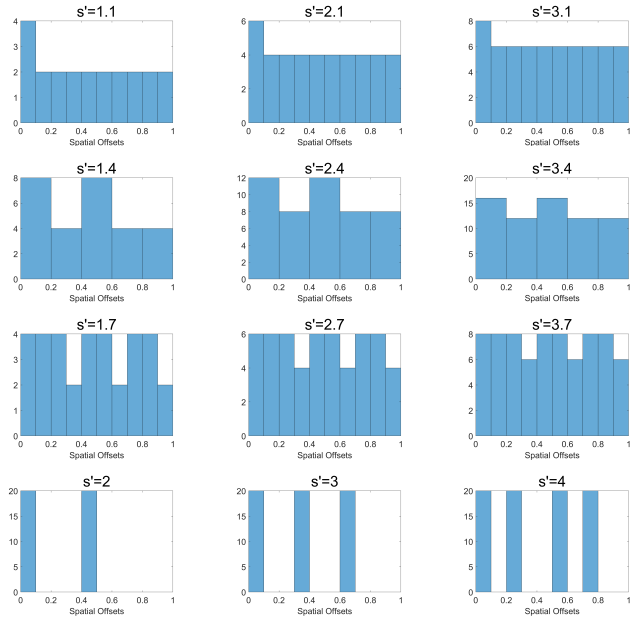


Fig. 6. Distributions of spatial offsets when the before-rescaling data contain 20×20 pixels.

as shown in Fig. 4), the spatial offsets of the pixels of the data with that target resolution are computed first, then the related spatial dynamic kernels are resolved via the mapping $f_o(\cdot; \mathbf{W}_0)$ and finally the rescaled data can be achieved with the convolution operation on the before-rescaling data. In our method, fully connected (FC) layers are jointly stacked as the kernel-generating module to transforms the spatial offset $\mathbf{v}$ into the dynamic kernel $\mathbf{k}$ (as shown in Fig. 3).

According to (4), the spatial offset v_x or v_y, i.e., the subpixel-level before-rescaling location of an after-rescaling pixel definitely lies on the interval $[0, 1)$. Fig. 6 graphically illustrates the histograms of x-direction spatial offsets under various zoom scale s' when the size of the before-rescaling image is 20×20. As can be observed, although spatial offsets distribute differently under different scales, especially among non-integer scales within two consecutive integer scales, their distributions are across the span $[0, 1)$, especially for non-integer scales. Intuitively, we can use data associated with one or very limited number of scales to train the offset-kernel mapping (i.e., the kernel generation model as shown in Fig. 4) along the offset span $[0, 1)$ as $f_o(\cdot; \mathbf{W}_0)$, which can then be generalized to yield data under other scales whose connected spatial offsets are still on the span $[0, 1)$.

More formally, once the kernel generation model $f_o(\cdot; \mathbf{W}_0)$ is obtained with the training using data associated with one or very limited number of scales, the model can in a testing phase be used to infer spatial dynamic kernels for any given spatial offsets, which could be yielded under any scales but always lie on $[0, 1)$, as shown in Fig. 4. The dynamic kernels are then utilized in the subsequent band-wise convolution step to generate spatial resolution-adjusted data. It is worth to note that, even if a testing scale connected to produce an ARHS image in the testing phase is far beyond those scales where the training process is performed on, its related spatial offsets still fall into the span $[0, 1)$ and, hence, the trained mapping model $f_o(\cdot; \mathbf{W}_0)$ still works for that un-predefined testing scale. This means that the kernel generation model trained on just a very limited number of zoom scales or even one scale can be generalized to yield dynamic kernels for rescaling data under any scales. This mechanism enables our method to yield data with arbitrary resolutions under a single trained model.

b) Band-wise dynamic convolution: As discussed above, dynamic kernels adaptively vary with spatial offsets which are determined by the specific spatial resolution-adjusting requirements. After kernel generation, the yielded kernels should be used in a convolution operation. The convolution in our method can be mathematically represented as

$$\widehat{\mathbf{H}}_{\mathrm{UP}}(i, j) = \mathbf{H}_{\mathrm{FEA}}\left(\left\lfloor \frac{i}{s'} \right\rfloor, \left\lfloor \frac{j}{s'} \right\rfloor\right) \otimes [\mathbf{k}_1(i, j), \ldots, \mathbf{k}_B(i, j)] \tag{6}$$

where $\widehat{\mathbf{H}}_{\mathrm{UP}}(i, j)$ represents the pixel (i, j) on the upsampled features $\widehat{\mathbf{H}}_{\mathrm{UP}}$, $\otimes$ denotes band-wise dynamic convolution around the $k \times k$ area centered at $(\lfloor \frac{i}{s'} \rfloor, \lfloor \frac{j}{s'} \rfloor)$ on $\mathbf{H}_{\mathrm{FEA}}$, and B is the band number of $\mathbf{H}_{\mathrm{FEA}}$. To avoid destroying the spectral information contained in $\mathbf{H}_{\mathrm{FEA}}$, the band-wise convolution is employed herein. In addition, band-wise dynamic convolution can reduce the required number of parameters compared with the conventional CNN convolution. For example, if both the number of bands of the input and output data are C and the spatial size of the kernels is $k \times k$, then, to generate a pixel with C-bands, our band-wise dynamic convolution requires $k \times k \times C$ parameters, while the conventional CNN convolution needs $k \times k \times C \times C$ parameters.

2) Spectral Augmentation: After spatially dynamic convolution, the yielded data is with the target arbitrary spatial resolution but with the same number of bands as the mediate spectral-spatial feature $\mathbf{H}_{\mathrm{FEA}}$. Moreover, as spatially dynamic convolution focuses on resizing the data along the spatial dimension, the raw spectral patterns of the input data may inevitably be distorted. To solve these issues, a spectral augmentation module f_{Spe} is applied succeeding to strengthen the recovery ability of our CNN for spectral patterns while adjust the data to the same number of bands as the input LRHS image. As depicted in Fig. 3, the module consists of serialized convolution layers, each of which is connected to kernels with spatial size 1×1. Intuitively, performing convolution along only one channel dimension enables the network to force spectral recovery without disturbance from the spatial neighborhood.

D. Composite Loss

In addition to the architecture of ARHS-CNN, its loss function is also crucial for the quality of the pansharpened ARHS image, which forces the network update and drives the network learning during the train phase (as shown in Fig. 4). Considering the proposed network architecture and task-specific demands, we adopt a novel composite loss function which involves a reconstruction constraint $\mathcal{L}_{\mathrm{recon}}$, a multistage constraint $\mathcal{L}_{\mathrm{stage}}$, and a spatial constraint $\mathcal{L}_{\mathrm{spatial}}$. The overall

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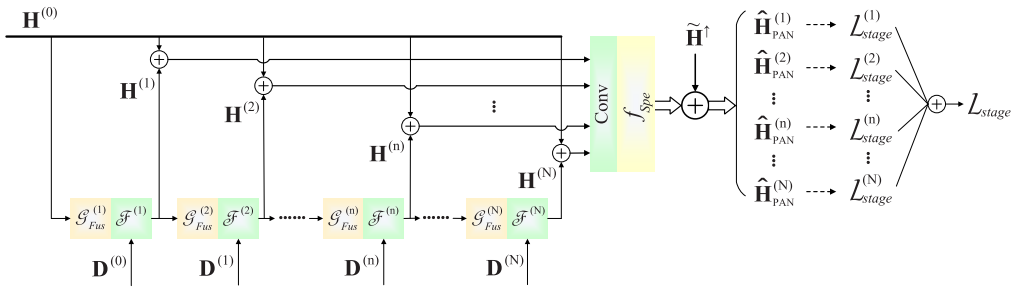


Fig. 7. Graphical illustration of the multistage constraint.

loss function is therefore defined as

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \alpha_1 \mathcal{L}_{\text{stage}} + \alpha_2 \mathcal{L}_{\text{spatial}} \tag{7}$$

where α_1 and α_2 denote the trade-off parameters of different constraints.

The reconstruction constraint $\mathcal{L}_{\text{recon}}$ is intended to measure the approximation of the reconstructed training data to the real training data. It forces the reconstructed high-resolution HS data $\widehat{\mathbf{H}}_{\text{TR}}$ to approximate the real high-resolution HS data $\mathbf{H}_{\text{TR}}$, which includes $sh \times sw$ pixels of each spectral band and b pixels of each spatial position. In addition, the pansharpened HS image with the PAN resolution, i.e., $\widehat{\mathbf{H}}_{\text{TR}}^{\downarrow s'}$ (that is degenerated from the reconstructed high-resolution HS data $\widehat{\mathbf{H}}_{\text{TR}}$) should approximate the standard HS pansharpening reference $\mathbf{H}_{\text{P}}$. Therefore, the reconstruction constraint is mathematically defined as

$$\mathcal{L}_{\text{recon}} = \|\widehat{\mathbf{H}}_{\text{TR}} - \mathbf{H}_{\text{TR}}\|_1 + \gamma \left\|\widehat{\mathbf{H}}_{\text{TR}}^{\downarrow s'} - \mathbf{H}_{\text{P}}\right\|_1 \tag{8}$$

where γ is a trade-off parameter that tunes the weights between two kinds of scale constraints. The degradation we apply here is based on the bicubic model.

The multistage constraint $\mathcal{L}_{\text{stage}}$ is based on the assumption that spatial-spectral features from different levels contribute to the final ARHS image in a coarse-to-fine fashion. When those features are processed by spectral augmentation, the corresponding outputs $\widehat{\mathbf{H}}_{\text{P}}^{(n)}$ are supposed to approach the ideal HS panshrpening image $\mathbf{H}_{\text{P}}$ with different quality levels. Generally, the final ARHS image relies more on the highest level features. Therefore, different weights are set for the constraints at different levels, and the weight increases as the network goes deeper, as graphically illustrated in Fig. 7. The multistage constraint is defined as

$$\mathcal{L}_{\text{stage}} = \sum_n \beta_n \left\|\widehat{\mathbf{H}}_{\text{P}}^{(n)} - \mathbf{H}_{\text{P}}\right\|_1, \quad \text{s.t. } \beta_n = \frac{n}{\sum_{n=1}^{N} n} \tag{9}$$

where $\widehat{\mathbf{H}}_{\text{P}}^{(n)}$ is the output of the nth RRDB group, followed by a convolution layer and f_{Spe} sequentially, and β_n weighs the contribution of different features (assuming that the last one deserves greater emphasis). From another perspective, the previous stage loss acts as a type of regularization for the parameters from the bottom layers.

The spatial constraint $\mathcal{L}_{\text{spatial}}$ is used to minimize the gradient distance between the reconstructed training HS data $\widehat{\mathbf{H}}_{\text{TR}}$ and the real training HS data $\mathbf{H}_{\text{TR}}$, which can drive the network to be more sensitive to spatial structure details. It can be formulated as

$$\mathcal{L}_{\text{spatial}} = \|\text{Grad}(\widehat{\mathbf{H}}_{\text{TR}}) - \text{Grad}(\mathbf{H}_{\text{TR}})\|_1 \tag{10}$$

where $\text{Grad}(\cdot)$ refers to the gradient maps obtained using (2).

It is worth noting that, as shown in (7), the L_1 norm is used in the design of our hybrid loss, rather than the L_2 norm often adopted in normal image enhancement tasks [28]–[30], [32], [33], which tends to incur spatial blurring [60]. Some of existing studies have revealed that the L_1 norm could achieve a better performance in image enhancement tasks [31], [58], [61].

After trained with the composite loss shown in (7), in the testing phase our ARHS-CNN can produce the ARHS image $\widehat{\mathbf{H}}_{\text{AR}}$ for a given LRHS image and the connected PAN, as shown in Fig. 4.

E. Summary

Our ARHS-CNN basically is composed of a multilevel enhancement subnetwork and a rescaling subnetwork (as shown by its overall architecture illustrated in Fig. 1), which corresponds to the two-step relay optimization process given in (1). The multilevel enhancement subnetwork incorporates the input LRHS and PAN images to yield intermediate spectral–spatial features with the same resolution as the PAN image, which is beneficial to fully and directly utilize the spatial structures of the PAN image and thus relieve the burden of computing intermediate features, tending to avoid spectral interference. In the rescaling subnetwork, an offset-kernel mapping mechanism is built to generate spatially dynamic kernels and, hence, fulfill the spatial resolution adjustment with a band-wise convolution, which are followed a spectral augmentation operation to tune the final number of the spectral bands and recover the additional spectral distortion of the reconstructed HS data.

The fundamental principle of our ARHS-CNN can be better unveiled by Fig. 4 which shows the data flow diagram for training and testing phases of the network. During the training phase of ARHS-CNN, multilevel enhancement subnetwork, kernel generation module, and spectral augmentation module are local networks updated under the supervising of the training data and the composite loss. Once the training process is accomplished, the obtained multilevel enhancement subnetwork model, kernel generation model, and spectral augmentation model will be transferred to be used in the testing

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TABLE I
PROPERTIES OF THE DATASETS USED FOR EVALUATION

Datasets	Data Size	Spectral Range	Training Scene	Testing Scene	Instrument
Washington DC Mall	PAN 1200×300 HS $240 \times 60 \times 210$	0.4–2.4 μm	PAN 1200×100 HS $240 \times 20 \times 210$	PAN 200×200 HS $40 \times 40 \times 210$	HYDICE
Pavia University	PAN 610×340 HS $122 \times 68 \times 115$	0.4–0.9 μm	PAN 610×140 HS $122 \times 28 \times 115$	PAN 200×200 HS $40 \times 40 \times 115$	ROSIS
Salinas Scene	PAN 510×215 HS $102 \times 43 \times 224$	0.4–2.5 μm	PAN 510×90 HS $102 \times 18 \times 224$	PAN 200×200 HS $40 \times 40 \times 224$	AVIRIS
Halls Creek	PAN 3483×567 HS $1161 \times 189 \times 171$	0.4–2.5 μm	PAN 3000×300 HS $1000 \times 100 \times 171$	PAN 360×360 HS $120 \times 120 \times 171$	Hyperion and ALI

phase to yield intermediate spectral-spatial features, spatially dynamic kernels and strengthened spectral prediction, and accordingly achieve the final ARHS image for the testing data. In the training phase, kernel generation model associated with a spatial offset-kernel mapping is determined under training zoom scales. As spatial offsets always lie on the interval [0, 1), the trained spatial offset-kernel mapping can be generalized to the testing phase, where no matter how the target resolution varies, its connected spatial offsets always lie on the fixed span [0, 1). On the other hand, our carefully designed multilevel enhancement subnetwork, spectral augmentation module and composite loss work collaboratively to relieve the spectral distortion during the spatial resolution adjustment. As a result, our ARHS-CNN can achieve HS images with any spatial resolution under a single trained CNN model, while keeping the spectral fidelity at any resolutions. In summary, our ARHS-CNN is able to handle the challenges involved in the new ARHS pansharpening task, which were presented in Section II. Algorithm 1 provides a detailed implementation of our ARHS-CNN.

Algorithm 1 ARHS-CNN

Input: Hyperspectral training dataset, $\mathbf{H}_{\mathrm{TR}}$; a set of selected training resolution ratios, $\mathbf{s}_{train}$; LR test HS image $\widetilde{\mathbf{H}}$ and its panchromatic counterpart $\mathbf{P}$; a user-customized resolution ratio, s_{test}

Output: HS image with the user-customized spatial resolution, $\widehat{\mathbf{H}}_{\mathrm{AR}}$

Randomly initialize model parameters $\boldsymbol{\theta_1}$ and $\boldsymbol{\theta_2}$;
Synthesize the low-resolution HS images and the corresponding PAN images by degrading $\mathbf{H}_{\mathrm{TR}}$ according to Wald's protocol;
while *not done* **do**
  Sample LR-HR batch of $s_i \in \mathbf{s}_{train}$;
  for *n epochs* **do**
    Obtain $\mathbf{H}_{\mathrm{FEA}}$;
    Calculate spatial offsets of all data pixels in the image after rescaling using (4);
    Generate dynamic kernels $\mathbf{k}$ using (5);
    Obtain the coarse rescaled data using (6);
    Apply spectral augmentation;
    Evaluate $\mathcal{L}$ using (7);
    Update $\boldsymbol{\theta_1}$ and $\boldsymbol{\theta_2}$ with respect to $\mathcal{L}$;
  end
end
Obtain the optimized model $\mathcal{M}_{\boldsymbol{\theta_1}}(\cdot)$ and $\mathcal{U}_{\boldsymbol{\theta_2}}(\cdot)$;
return $\widehat{\mathbf{H}}_{\mathrm{AR}} = \mathcal{U}_{\boldsymbol{\theta_2}}(\mathcal{M}_{\boldsymbol{\theta_1}}(\widetilde{\mathbf{H}}, \mathbf{P}), s_{test})$

IV. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we present the experimental datasets and numeric metrics applied for performance evaluation, and then describe some details of the experimentation with our ARHS-CNN method, including the model hyperparameters and training settings. Finally, the high-quality images from different methods are compared through both quantitative and qualitative evaluations.

A. Experimental Datasets and Quantitative Metrics

Our newly proposed ARHS-CNN method is tested on four widely used HS datasets, whose major properties are listed in Table I.

1) Washington DC Mall dataset. Collected by the hyperspectral digital imagery collection experiment (HYDICE) sensor, it covers an urban area in Washington DC Mall. The spectral range spans from 0.4 to 2.4 μm with 210 bands. About 191 bands were used in our experiments, while low SNR bands were removed.
2) Pavia University dataset. Yielded by the reflective optics system imaging (ROSIS) sensor, it covers an urban area in Pavia, northern Italy. Its spectral range is from 0.4 to 0.9 μm formed with 115 bands. About 103 bands were used in our experiments.
3) Salinas Scene dataset. Obtained by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor, it covers a rural area in Salinas Valley, California. It spectrally covers a range from 0.4 to 2.5 μm using 224 bands. About 204 bands were used in our experiments.
4) Halls Creek dataset. Contains an HS image and a real PAN image, which cover the same land area. They are acquired by Hyperion sensor and advanced land imager (ALI) from Earth Observing-1 (EO-1) satellite, respectively. The covered area is a vegetative area in Halls Creek, Australia. The spectral range of the HS

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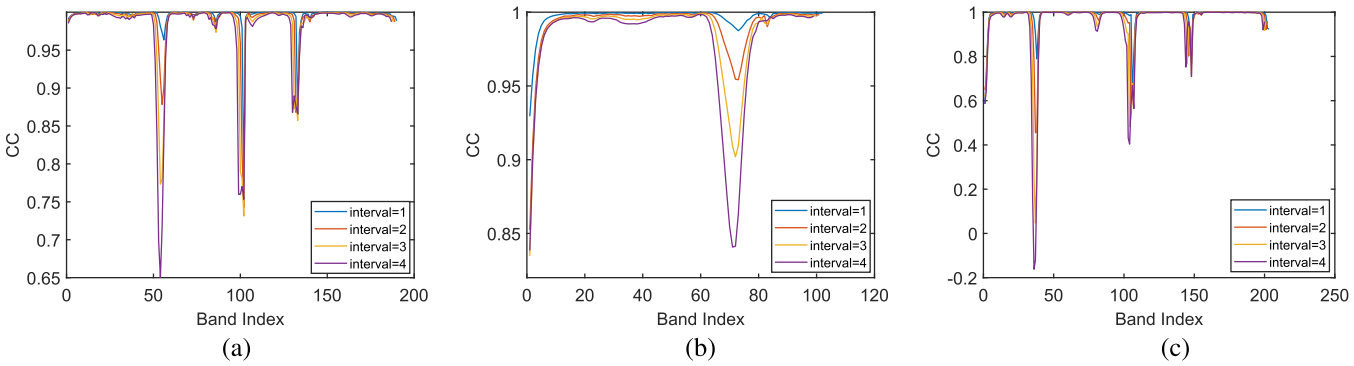


Fig. 8. CCs of different datasets (with different intervals). (a) Washington DC Mall. (b) Pavia university. (c) Salinas scene.

image spans from 0.4 to 2.5 μm using 242 bands, where 171 bands were used in our experiments.

It is noteworthy that the quantitative evaluation of HS pansharpening is relatively complicated owing to a shortage of real references [2]. Employing semi-synthetic HS and PAN images is an alternative strategy. In this case, the reference image, i.e., ground-truth (GT), is the observed HS image in real world, whereas the LRHS and PAN images which need to be integrated are synthesized from GT by Wald's protocol [62].

Supported by the reference image and the synthesized data, quantitative metrics can be utilized to perform methods comparison. Here, we take into consideration some of the most prevail and widely-used metrics to fulfill evaluations [2], [34]–[36], [63]: peak signal-to-noise ratio (PSNR) [24], structure similarity index (SSIM) [64], cross correlation (CC) [2], spectral angle mapper (SAM) [65], root mean squared error (RMSE) [2], Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) [66], and an extension of the universal image quality index (Q2^n) [67]. CC is a spatial index and SAM is a spectral index, while the rest four serve as global indexes. For the real HS dataset, the Halls Creek dataset, we additionally apply three metrics in full-resolution validation, i.e., D_λ, D_s and QNR [68].

B. Implementation Details

In this section, we mainly illustrate the key settings of our proposed ARHS-CNN, including model hyperparameters and training settings.

All feature interaction layers in the multilevel enhancement subnetwork, i.e., $\mathcal{F}^{(n)}$, consist of a concatenation operation and a convolution layer with 64 filters, whose spatial size is generally 3×3. The activation function we adopted is the leaky rectified linear activation with 0.2 slope on negative inputs. It should be noted that the spectral augmentation module f_{Spe} shares its weights when calculating the multistage constraint L_{stage}. pBesides, RRDBs in our network are set empirically. Each of the RRDB group $\mathcal{G}^{(n)}_{\text{Fus}}$ and the RRDB group $\mathcal{G}^{(n)}_{\text{Grad}}$ includes two RRDBs, whose settings follow [56]. The RRDB group $\mathcal{G}^{(n)}_{\text{Fus}}$ and the RRDB group $\mathcal{G}^{(n)}_{\text{Grad}}$ are used in pairs, and we totally employ three pairs of the groups. Another important hyperparameter is the size of the 3-D convolution kernels along the spectral dimension in our IF module (i.e., c shown in Fig. 2), which is determined by the correlation among neighboring bands. Fig. 8 exhibits correlation coefficients between those bands with a certain interval from different datasets. It can be noticed that the CC decreased as the interval raises and vice-versa, which demonstrates that neighboring bands are highly correlated. By jointly considering computational complexity and performance, c is set empirically to 3, and the convolution stride is defined as two without padding.

As regards the training procedure, the real observed HS image serves as the ARHS pansharpening reference image $\mathbf{H}_{\text{TR}}$, while the standard HS pansharpening reference $\mathbf{H}_{\text{P}}$ is yielded by Gaussian blurring and downsampling with ratio s'. Meanwhile, the LRHS image $\widetilde{\mathbf{H}}$ is generated by a similar degradation process with ratio $r = 5$, and the connected PAN image is produced by the average of the visible bands of $\mathbf{H}_{\text{P}}$. Furthermore, we spatially pre-interpolate $\widetilde{\mathbf{H}}$ to the same size as the PAN with bicubic interpolation, yielding $\widetilde{\mathbf{H}}^{\uparrow}$. As the offset-kernel mapping mechanism, we select a finite set of s' (from 1 to 2.5 with a stride of 0.1). During the training, s' is randomly picked for each batch, and then four LR patches with size of 20×20 pixels are randomly cropped. Data augmentation is performed through random rotation and random flipping when necessary.

An Adam optimizer, with the default setting $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\varepsilon = 10^{-8}$, is used to train the proposed ARHS-CNN [69], where the initial learning rate is empirically chosen to be 1×10^{-4}. Learning rate is adjusted during the training in terms of a exponential decay. The total iterations is set to 1 00 000 and the learning rate would get to 1×10^{-5} when the iteration process ends. Our experiments have been ran with the Tensorflow deep learning platform in an environment comprising an Intel i5-8400 CPU, 16-GB RAM, and 6-GB NVIDIA GTX 1060 graphics processing unit.

C. Arbitrary Resolution Pansharpening

Our ARHS panshening CNN is able to achieve pansharpening with an arbitrary resolution, including the PAN resolution which is connected to the standard HS pansharpening. We first focus on the general ARHS panshening case, then the standard HS pansharpening.

Nine state-of-the-art HS pansharpening methods are involved in those experiments, including Gram–Schmidt adaptive (GSA) [10], CNMF [24], hyperspectral super-resolution (HySure) [20], Bayesian sparsity promoted prior (BayeSparse) [19], detail-based deep Laplacian pansharpening (DDLPS) [34], hyperspectral pansharpening via

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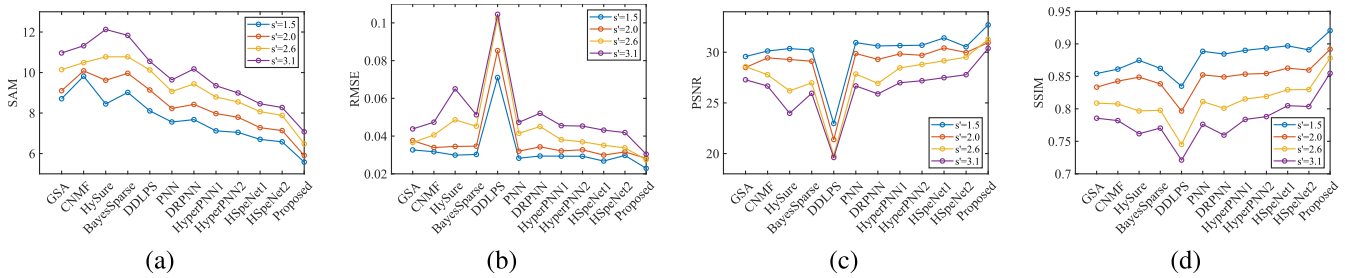


Fig. 9. Pansharpening performances under arbitrary s' on Washington DC Mall dataset. (a) SAM. (b) RMSE. (c) PSNR. (d) SSIM.

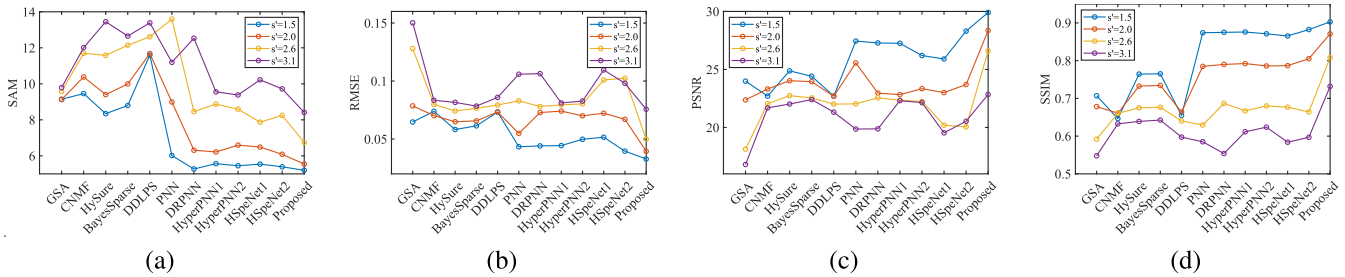


Fig. 10. Pansharpening performances under arbitrary s' on Pavia University dataset. (a) SAM. (b) RMSE. (c) PSNR. (d) SSIM.

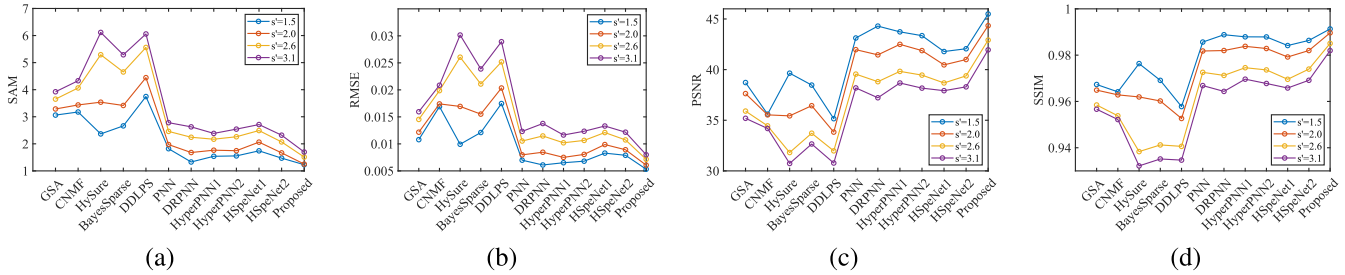


Fig. 11. Pansharpening performances under arbitrary s' on Salinas Scene dataset. (a) SAM. (b) RMSE. (c) PSNR. (d) SSIM.

spectrally predictive CNNs (HyperPNNs) [33], and hyperspectral spectral-fidelity networks (HSpeNets) [38]. Besides, two representative CNNs, i.e., PNN [28], and deep residual pan-sharpening neural network (DRPNN) [29], which were originally designed for MS pansharpening were fulfilled for the HS case. Since they have never tackled the issue of arbitrary resolution, in the ARHS pansharpening experiments we boost existing standard pansharpening methods by attaching an additional bicubic interpolation as the last layer for comparison. In addition, GSA, CNMF, HySure and BayesSparse are implemented according to the code provided by [2], and DDLPS is reproduced based on the code downloaded from the home page of the authors, without modifications of parameters. For CNN-based methods, their parameters settings strictly follow the descriptions in the connected publications. For a relatively fair comparison, we apply the same optimizer setting to all those CNN-based methods, i.e., an Adam optimizer with the default setting, and the initial learning rates of those methods are all set to 1×10^{-4}. As regards training sample setting, they are also under the same one: spatial size of each patch is set to 15×15 and the total number of the training samples is 2560.

Figs. 9–11 illustrate the numeric results of ARHS pansharpening from Washington DC Mall, Pavia University, and Salinas Scene datasets, respectively, and Figs. 12–14 display the graphical results. First, for quantitative assessments, we randomly take 1.5, 2, 2.6, and 3.1 as the user-customized scales. The experimental results on Washington DC Mall, Pavia University and Salinas Scene datasets are demonstrated in Figs. 9–11, respectively. Herein, we take the same HS image as reference and degrade it with the user-customized scales, yielding different LRHS images as the CNN inputs. The outputs of different methods and different scales are finally evaluated by the selected metrics. As can be observed, our newly proposed ARHS-CNN achieves the best results for all the four resolutions, gaining better SAM, RMSE, PSNR, and SSIM performances than the others, especially when the scale becomes larger. It can be noticed that the same method may have inconsistent performances on different datasets, especially for those conventional methods, such as HySure and DDLPS. This is blame for the mismatching between the predefined model parameters and the dataset characteristics, while CNN-based methods achieve relatively better metrics than those conventional ones. From Figs. 9 to 11, it can also be observed that, with the rising of the user-customized scale (i.e., the target pansharpening resolution), SAM, RMSE, PSNR, and SSIM performances tend to decline. This is in line with the intuitiveness that the higher target spatial resolution implies the more difficult the ARHS pansharpening is.

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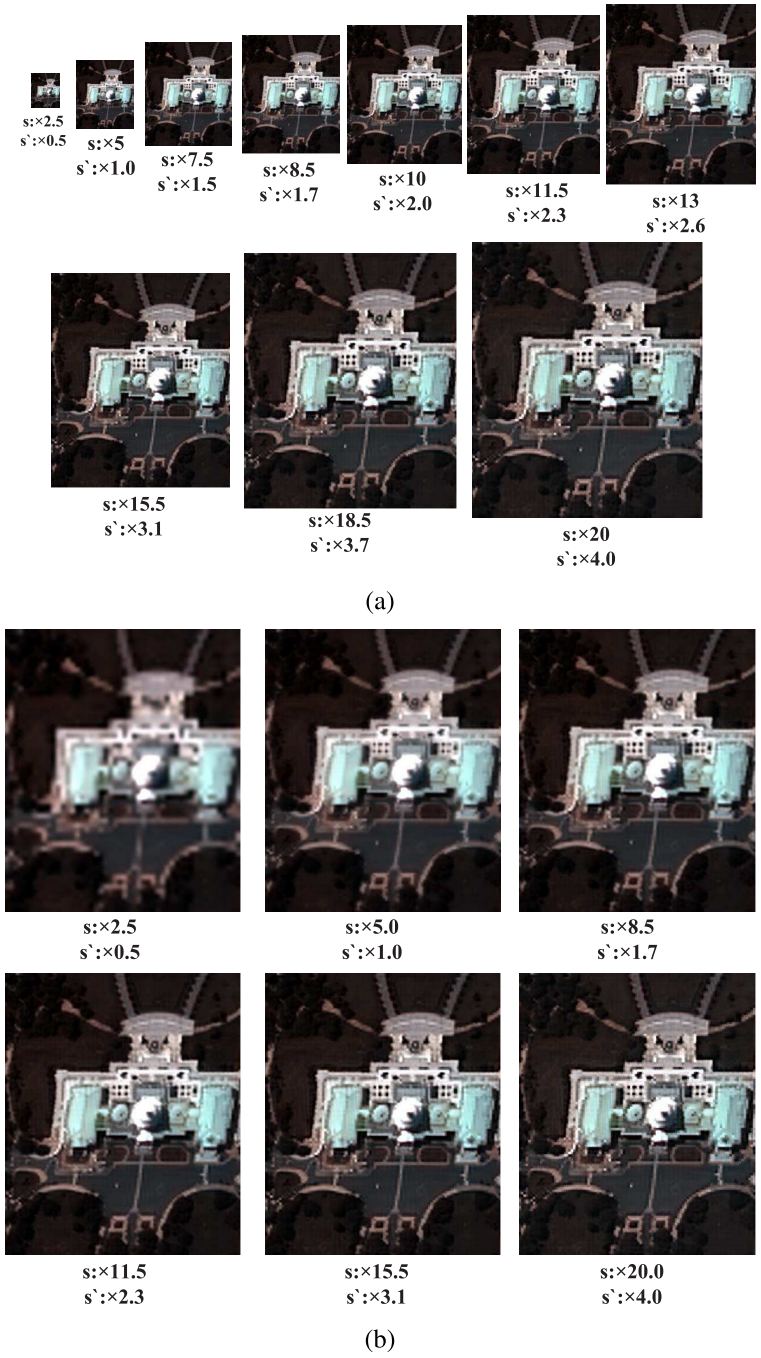


Fig. 12. Visual effects of our ARHS-CNN, on a sub-scene of Washington DC Mall dataset. (a) Original ARHS pansharpening results. (b) Upscale versions of the results with the same size.

Second, for a qualitative and intuitive comparison, we visualize the ARHS pansharpening results of ARHS-CNN in Figs. 12(a), 13(a), and 14(a), where ARHS pansharpening scales within and beyond the training range are involved. Different from the previous quantitative assessment, in the test phase, ARHS-CNN (with frozen parameters) takes the same LRHS and PAN images as inputs, and outputs different ARHS images according to the user-customized scales. Here, we choose different scales s for testing, ranging from 2.5 to 20. It is notorious that the demand for detail reconstruction is growing as the scale enlarged, so that the task of ARHS pansharpening becomes more and more difficult. Generally, the spatial resolution gap between the LRHS and HRHS images is range from 3 to 8 in most of the existing pansharpening methods, based on the observed PAN image. However, our ARHS-CNN is able to generate HS images with any spatial resolution under a single trained model, including those resolutions much higher than LRHS, as illustrated results of $s = 20$ shown in Figs. 12(a), 13(a), and 14(a). For the purpose of more direct comparison, we also display images with different scales in the same size by interpolation [as shown in Fig. 12(b), Fig 13(b), and Fig 14(b)]. Indeed, our

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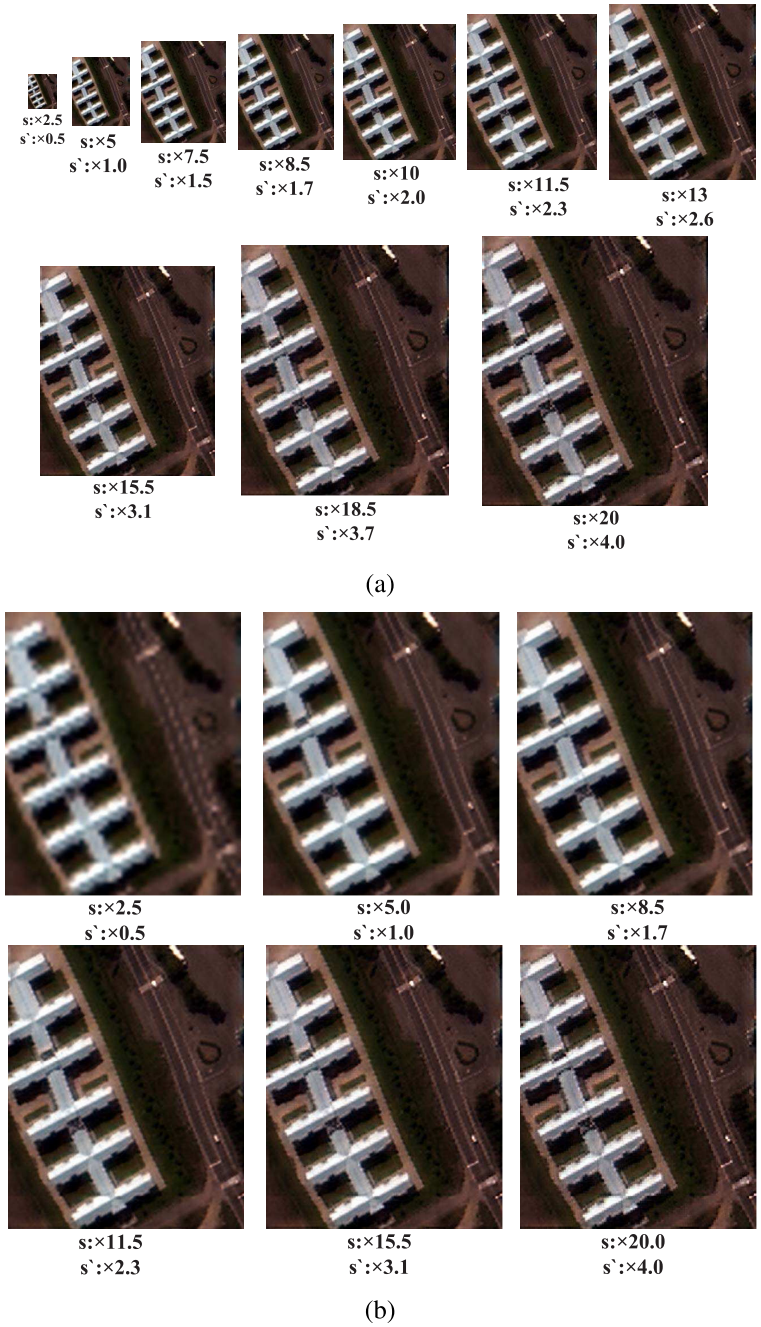


Fig. 13. Visual effects of our ARHS-CNN, on a sub-scene of Pavia University dataset. (a) Original ARHS pansharpening results. (b) Upscale versions of the results with the same size.

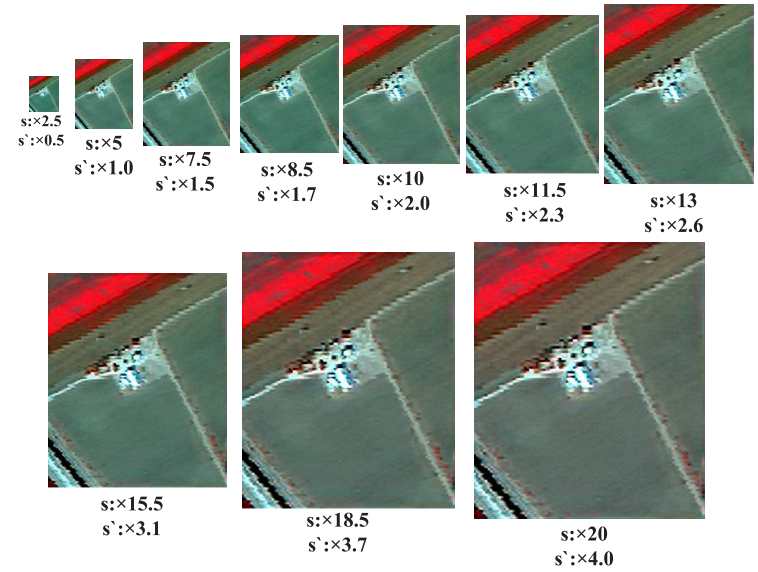
ARHS-CNN is competent in the task of ARHS pansharpening, even with a large-scale demand. It is worth noting that spectral distortion tends to be a little more severe as the ARHS pansharpening scale increases for the Salinas Scene dataset, as shown in Fig. 14. As the ARHS pansharpening scale increases, HS spatial details need to be reconstructed grow rapidly while the PAN spatial details that could help the reconstruction still keep unchanged, which may result in a tendency to incur more serious spectral distortion during enhancing the HS spatial resolution. With a careful design, our ARHS-CNN is able to effectively restrain the spectral distortion when improving LRHS images to an arbitrary spatial resolution.

Briefly speaking, our ARHS-CNN achieves superior performance on almost all scale factors, which truly verifies its effectiveness on ARHS pansharpening.

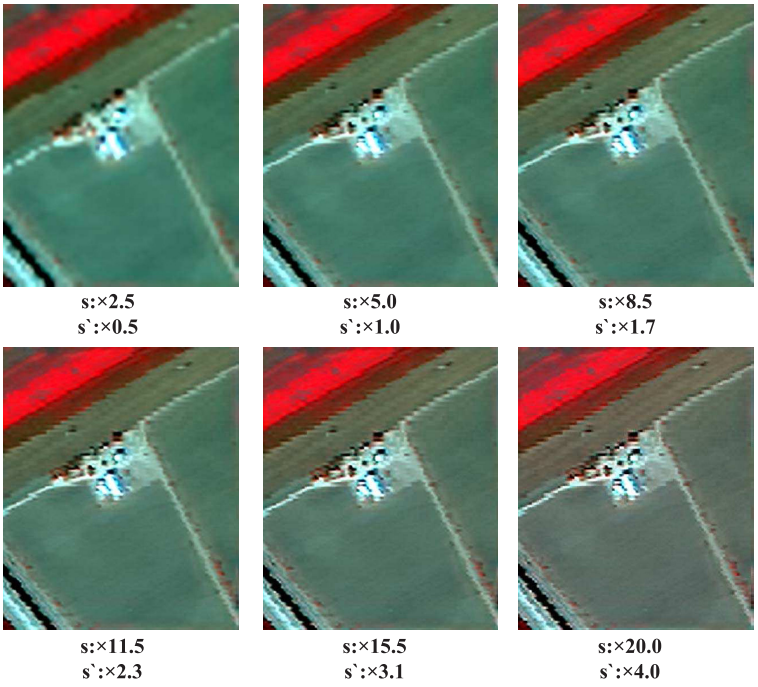
D. Standard Pansharpening

If the intermediate pansharpening scale s' is set to 1, i.e., the target ARHS pansharpening resolution equals to that of the PAN, the ARHS pansharpening is degenerated to the standard HS pansharpening. Tables II–IV

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(a)



(b)

Fig. 14. Visual effects of our ARHS-CNN, on a sub-scene of Salinas Scene dataset. (a) Original ARHS pansharpening results. (b) Upscale versions of the results with the same size.

TABLE II

NUMERIC RESULTS OF COMPARED METHODS ON WASHINGTON DC MALL DATASET

	GSA	CNMF	HySure	BayesSparse	DDLPS	PNN	DRPNN	HyperPNN1	HyperPNN2	HSpeNet1	HSpeNet2	Proposed
CC(↑)	0.9167	0.8861	0.9172	0.9297	0.8357	0.9423	0.9240	0.9287	0.9291	0.9435	0.9350	**0.9530**
SAM(↓)	6.7066	8.9472	6.9372	6.6885	8.8507	5.2270	5.0536	5.2651	5.3508	4.9850	5.2521	**4.6050**
RMSE(↓)	0.0161	0.0195	0.0158	0.0145	0.0215	0.0125	0.0129	0.0137	0.0136	0.0128	0.0138	**0.0113**
ERGAS(↓)	4.6597	5.5299	4.8869	4.3864	14.4725	4.1465	5.3677	4.6240	4.5262	3.9100	4.5004	**3.6274**
PSNR(↑)	35.7665	34.2620	35.9549	36.3970	33.3596	37.6877	37.5575	37.1431	37.1813	37.6314	37.1371	**38.9017**
SSIM(↑)	0.9360	0.9187	0.9385	0.9404	0.9061	0.9588	0.9576	0.9567	0.9563	0.9583	0.9567	**0.9667**
$Q2^n$(↑)	0.8937	0.8653	0.8875	0.9069	0.6566	0.9243	0.8946	0.9127	0.9142	0.9275	0.9186	**0.9394**

present the comparisons of numeric results on Washington DC Mall, Pavia University and Salinas Scene datasets, and Fig. 15 compares the graphical results of our ARHS-CNN with other standard pansharpening methods.

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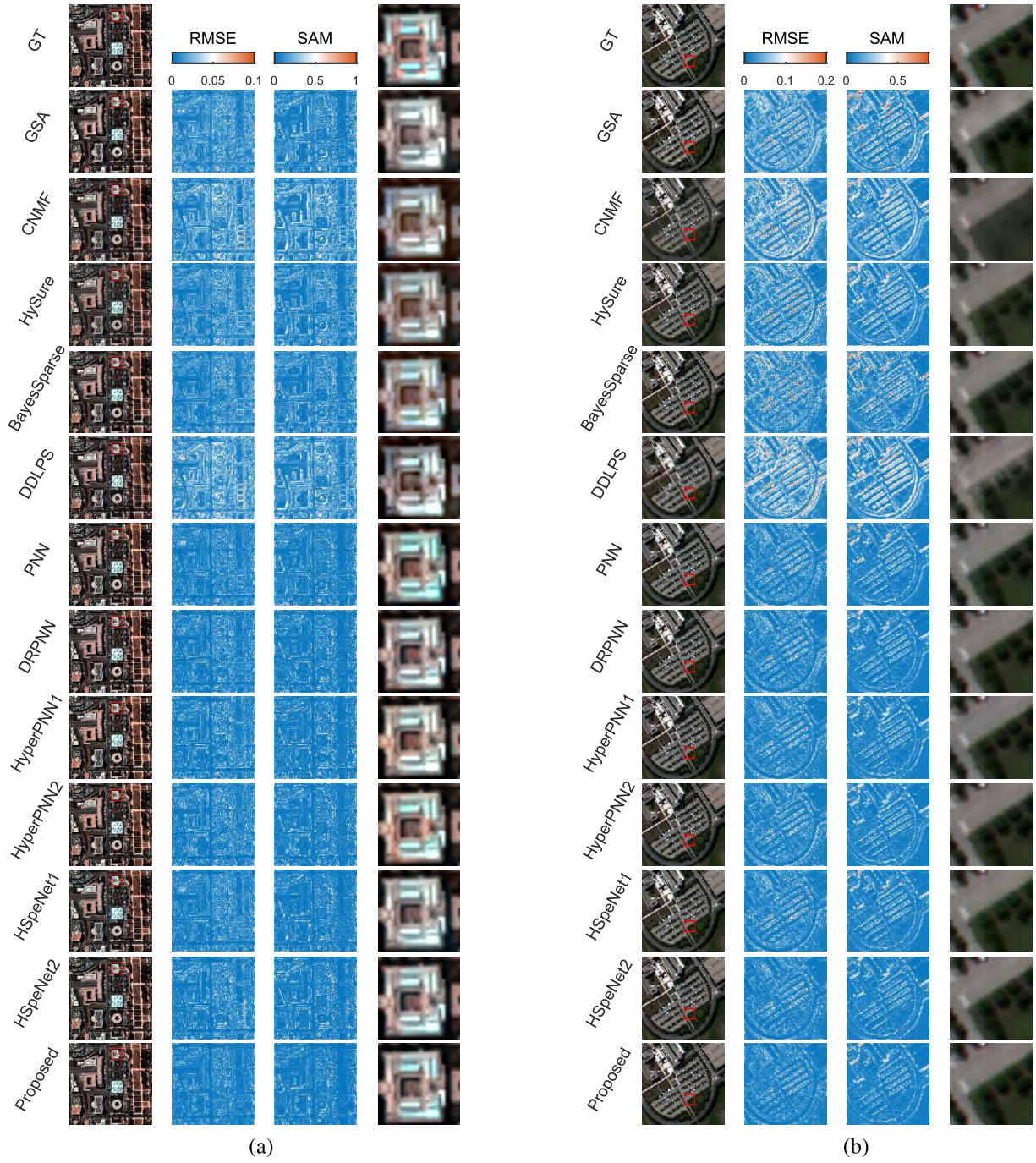


Fig. 15. Qualitative analysis on Washington DC Mall and Pavia University datasets. The first column shows the fusion results in form of color-composite. The second and third columns are the mean RMSE error and the SAM error of the HS cube, respectively; the fourth column is an enlarged area of the color-composite image. (a) Washington DC Mall dataset. (b) Pavia University dataset.

Consider the numeric results gathered in Tables II–IV, where the best results are highlighted in bold typeface. Among the conventional methods, BayesSparse provides the best performance on Washington DC Mall and Salinas Scene datasets, while slightly lags behind on Pavia University dataset. CNMF is the most inferior method, which suggests its model parameters needed to be updated for processing these datasets. As observed in these tables, methods based on CNN perform better than the conventional ones. Therein, the proposed ARHS-CNN gains significant advantage over the other methods under all the considered metrics, especially for SAM. The results implies that our ARHS-CNN is able to not only reconstruct real-world spatial details, but also reliably guarantee spectral fidelity. Besides, HSpeNets take the role of the closest competitors, with better scores in terms of SAM and PSNR. DDLPS, as a combination of CNN and model

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TABLE III
NUMERIC RESULTS OF COMPARED METHODS ON PAVIA UNIVERSITY DATASET

	GSA	CNMF	HySure	BayesSparse	DDLPS	PNN	DRPNN	HyperPNN1	HyperPNN2	HSpeNet1	HSpeNet2	Proposed
CC(↑)	0.8966	0.8198	0.8888	0.8951	0.8022	0.9519	0.9571	0.9582	0.9556	0.9537	0.9642	**0.9669**
SAM(↓)	8.2025	8.4263	6.9818	7.3495	9.1709	5.4915	4.8403	4.9798	5.0198	5.2249	4.7432	**4.6130**
RMSE(↓)	0.0475	0.0605	0.0509	0.0512	0.0618	0.0326	0.0310	0.0303	0.0313	0.0318	0.0280	**0.0268**
ERGAS(↓)	5.0857	5.8053	4.7505	4.6990	6.8550	3.1479	3.0114	2.9148	2.9958	3.0916	2.7110	**2.6468**
PSNR(↑)	26.5057	24.4962	25.9233	25.8149	24.1800	29.7489	30.1798	30.3801	30.0930	29.9467	31.0698	**31.3955**
SSIM(↑)	0.7682	0.7276	0.8127	0.8145	0.7160	0.8931	0.9051	0.9086	0.9040	0.9021	0.9196	**0.9201**
$Q2^n$(↑)	0.8579	0.7718	0.8618	0.8711	0.7493	0.9341	0.9388	0.9441	0.9404	0.9370	0.9507	**0.9531**

TABLE IV
NUMERIC RESULTS OF COMPARED METHODS ON SALINAS SCENE DATASET

	GSA	CNMF	HySure	BayesSparse	DDLPS	PNN	DRPNN	HyperPNN1	HyperPNN2	HSpeNet1	HSpeNet2	Proposed
CC(↑)	0.9567	0.9147	0.9576	0.9695	0.9195	0.9732	0.9704	0.9726	0.9743	0.9759	0.9729	**0.9820**
SAM(↓)	2.8061	2.8582	2.0940	1.7819	2.8626	1.7692	1.4386	1.4469	1.4482	1.4257	1.4129	**1.2327**
RMSE(↓)	0.0094	0.0164	0.0109	0.0080	0.0137	0.0068	0.0068	0.0061	0.0064	0.0066	0.0070	**0.0053**
ERGAS(↓)	2.2848	2.8439	2.0642	1.5514	5.6656	1.6518	2.1884	1.6721	1.7302	1.4343	1.6741	**1.2780**
PSNR(↑)	40.1056	35.7267	39.2527	41.9249	37.2674	43.3768	43.3304	44.2600	43.9244	43.5910	43.1465	**45.5146**
SSIM(↑)	0.9713	0.9669	0.9787	0.9821	0.9670	0.9883	0.9897	0.9906	0.9908	0.9898	0.9896	**0.9922**
$Q2^n$(↑)	0.8441	0.8229	0.8780	0.9110	0.6904	0.9096	0.9138	0.9216	0.9228	0.9217	0.9183	**0.9352**

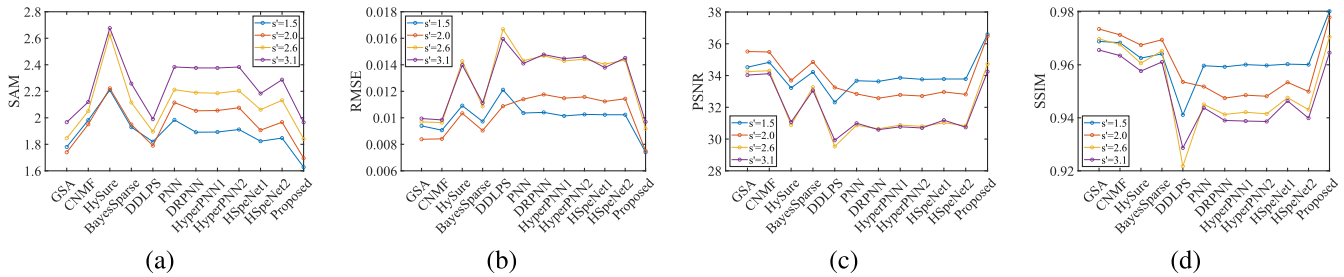


Fig. 16. Panshparpening performances under arbitrary s' on Halls Creek dataset. (a) SAM. (b) RMSE. (c) PSNR. (d) SSIM.

optimization, seems poor in standard pansharpening on all datasets, which may turn better by carefully finetuning its model parameters.

For a further comprehensive assessment of these methods, qualitative analysis is necessary. We first conduct visual inspections on the Washington DC Mall dataset. In Fig. 15(a), the first column illustrates the RGB color-composite images yielded from the pansharpened HS images, whereas the second column displays the RMSE heatmaps, indicating the pixel-wise reconstruction errors for the HS image. The third column of Fig. 15(a) is left for the SAM error heatmaps, which suggest the spectral fidelities of the reconstructed images, and the fourth column presents the enlarged sub-scenes of the color-composite images. In first column, the color-composite images of almost all the methods look high quality, while significant differences among the methods can be found in the rest columns. When the RMSE and SAM heatmaps are considered, the weaknesses of CNMF and DDLPS are apparent, while PNN and HSpeNets and our proposed ARHS-CNN achieve more closer images to the ground truth than the other methods. As for enlarged RGB sub-scenes of the pansharpening results, we can find that GSA incurs an under-saturated result while PNN triggers an over-saturated result, which implies that there are spectral distortions not negligible. Overall, HySure and BayesSparse achieve relatively good visual inspection results among conventional methods. For CNN-based methods, DRPNN and HyperPNNs suffer from subtle diffused blurring and PNN performs worse in spectral preserving. Our proposed ARHS-CNN method produces the best visual results, with relatively excellent spatial quality and without noticeable spectral distortions. Fig. 15(b) displays the visual outcome from the Pavia University dataset. Similar to the case of the first dataset, our ARHS-CNN also achieves excellent performance. The HS images yielded by GSA, HySure and BayeseSparse produce relatively good results, while CNMF and DDLPS undergo severe spectral distortion. For the methods based on CNN, DRPNN and HyperPNNs tolerate slight spatial blurring. The HS images yielded by PNN and HSpeNets are closer to the GT image, as implied by the RMSE and SAM heatmaps.

E. Real HS Dataset Validation

To make the validation more complete, experimental results on the real HS dataset, i.e., the Halls Creek dataset, are given here. Following the experimental pipeline above, we also first focus on the general ARHS panshening case, then the standard HS pansharpening.

The numeric results is demonstrated in Fig. 16. In the experiment, the user-customized scales are randomly set as 1.5, 2.0, 2.6, and 3.1. The outputs of different methods and different scales are finally evaluated by SAM, RMSE, PSNR, and

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TABLE V
NUMERIC RESULTS OF COMPARED METHODS ON HALLS CREEK DATASET

	GSA	CNMF	HySure	DDLPS	PNN	DRPNN	HyperPNN1	HyperPNN2	HSpeNet1	HSpeNet2	Proposed
CC(↑)	0.9369	0.8836	0.8926	0.9540	0.9116	0.9059	0.9266	0.9257	0.9608	0.9535	**0.9651**
SAM(↓)	0.9481	1.3109	1.3678	0.9346	1.3457	1.1653	1.1622	1.1901	0.8746	0.8991	**0.8479**
RMSE(↓)	0.0058	0.0062	0.0059	0.0045	0.0052	0.0048	0.0046	0.0048	0.0043	0.0043	**0.0042**
ERGAS(↓)	1.7858	2.5552	2.5200	3.0090	2.4917	3.1529	2.4729	2.3489	1.5361	1.9367	**1.5198**
PSNR(↑)	44.8057	44.2143	44.5343	40.8619	45.6012	46.4234	46.6655	46.4571	47.3082	47.4316	**47.5272**
SSIM(↑)	0.9845	0.9830	0.9834	0.9890	0.9876	0.9885	0.9897	0.9894	0.9908	0.9912	**0.9915**
$Q2^n$(↑)	0.8666	0.7923	0.7967	0.8846	0.8478	0.8373	0.8716	0.8703	0.9099	0.9071	**0.9171**
D_λ(↓)	0.0497	0.0920	0.0664	0.0621	0.0361	0.0300	0.0468	0.0513	0.0373	0.0294	**0.0289**
D_s(↓)	0.1908	0.2218	0.1615	0.1475	0.0581	0.0395	0.0244	0.0271	0.0429	0.0191	**0.0129**
QNR(↑)	0.7690	0.7067	0.7829	0.7996	0.9079	0.9317	0.9300	0.9229	0.9214	0.9521	**0.9585**

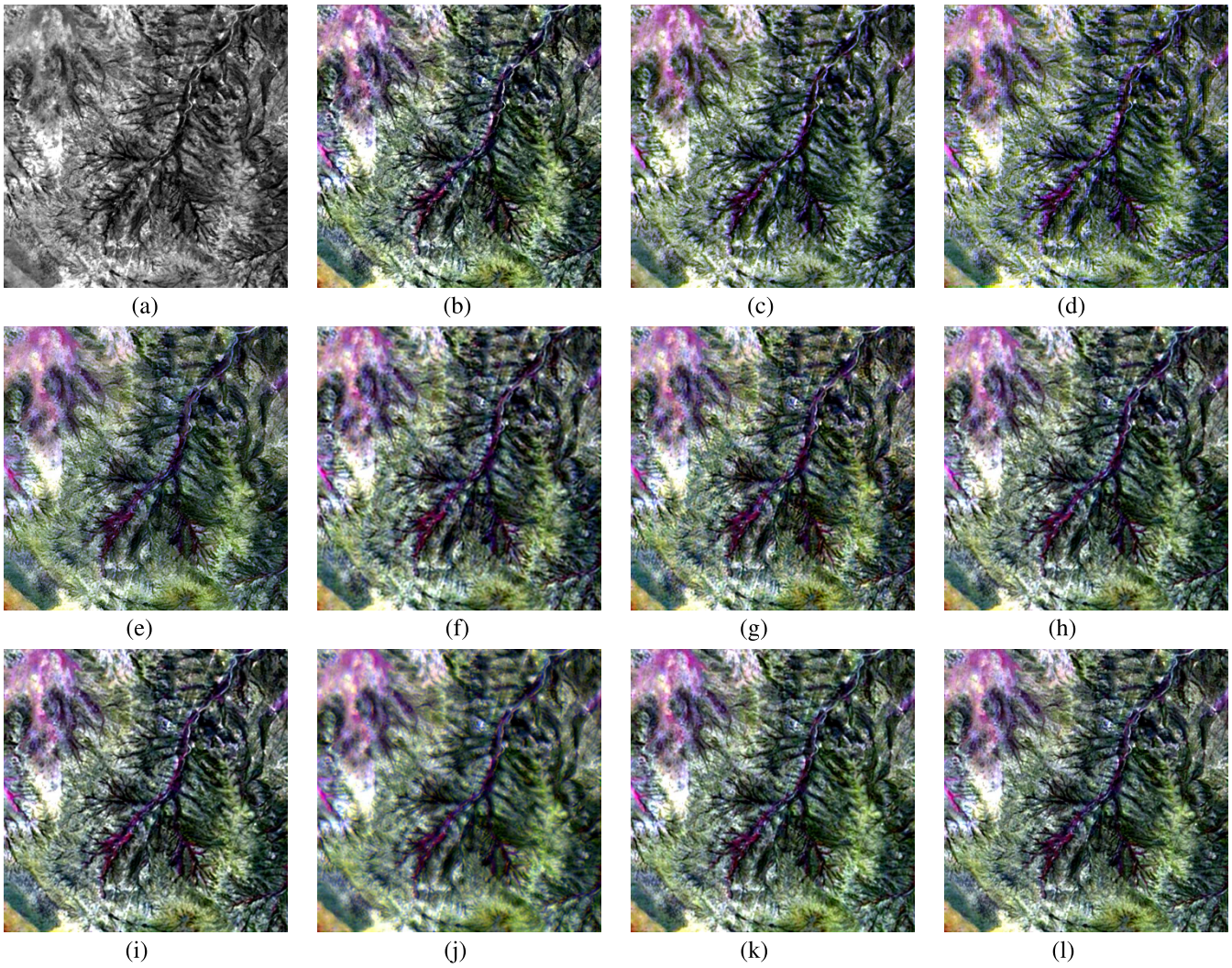


Fig. 17. Qualitative analysis on Halls Creek dataset. (a) PAN. (b) GSA. (c) CNMF. (d) HySure. (e) DDLPS. (f) PNN. (g) DRPNN. (h) HyperPNN1. (i) HyperPNN2. (j) HSpeNet1. (k) HSpeNet2. (l) Proposed.

SSIM. It can be observed that our ARHS-CNN achieves the best results for all the four resolutions, gaining better metrics scores than the others. Besides, the phenomenon that metrics tend to decline with the rising of the user-customized scale (i.e., the target pansharpening resolution), remains consistent as the other three datasets tested above.

As regards standard pansharpening testing on the Halls Creek dataset, we mainly concentrate on full-resolution validation. It is noteworthy that the BayesSparse [19] used for performance comparison on simulated HS datasets is excluded since the necessary blur matrix for its implementation procedure is unavailable. To perform the quantitative evaluation of this real HS dataset, we introduce consistency property [70] and the quality with no reference (QNR) index [68]. The former one assesses the similarity between the degraded pansharpened HS image and the originally observed HS image by means of CC, SAM, RMSE, ERGAS, PSNR, SSIM, and $Q2^n$. The latter one directly assesses the pansharpened at full-resolution level through D_λ, D_s, and QNR indexes [71]. Table V presents the comparisons of numeric results on the

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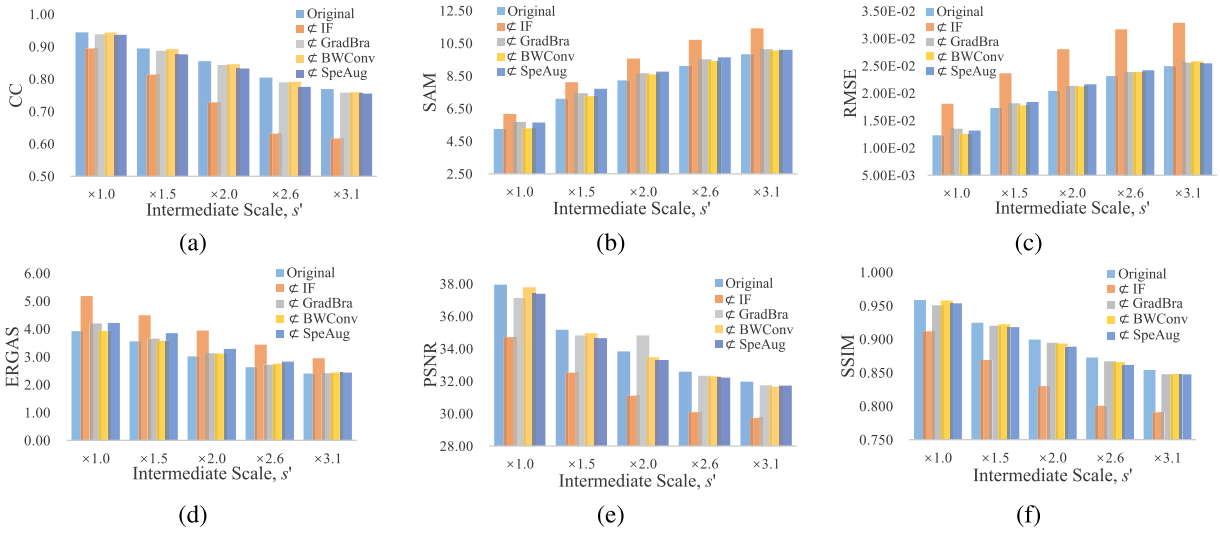


Fig. 18. Ablation study for the proposed network structure. $\not\subset$ means removing part of the module and studying its performances with different metrics. (a) CC. (b) SAM. (c) RMSE. (d) ERGAS (e) PSNR. (f) SSIM.

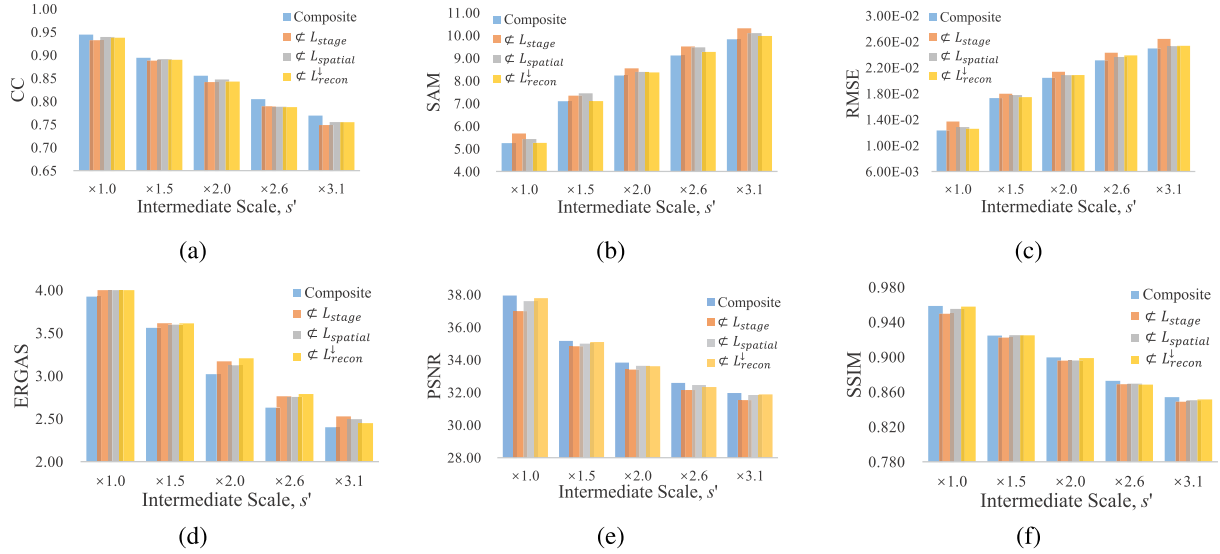


Fig. 19. Ablation study for the composite loss function. $\not\subset$ means removing part of the module and studying its performances with different metrics. (a) CC. (b) SAM. (c) RMSE. (d) ERGAS (e) PSNR. (f) SSIM.

Halls Creek dataset, and Fig. 17 compares the qualitative results of our ARHS-CNN with other standard pansharpening methods. By comparison, the experimental results shown in Table V and Fig. 17 fully demonstrate the superiority of ARHS-CNN in terms of both numeric results and visual appearance, which validates the outstanding overall performance of our proposed method for HS pansharpening.

F. Ablation Study

To further investigate the contributions from different modules in the proposed ARHS-CNN, we implement some ablation studies on the Washington DC Mall dataset. In this experiment, we can clearly see the performances degradation when removing certain parts of the network structure or loss function.

In consideration of network structure, we mainly explore the effectiveness of the modules including IF, gradient branch (GradBra), band-wise convolution (BWConv), and spectral augmentation (SpeAug). As shown in Fig. 18, $\not\subset$ means removing certain parts of the network structure. Noticeably, the performance suddenly drops when removing the IF module, indicating that the specialized preprocessing is one of the most essential parts of the whole ARHS pansharpening. Compared to this module, removing one of the other branches interacts only with less effect. Moreover, the spectral augmentation module does guarantee spectral fidelity at any target spatial resolution, shown in Fig. 18(b). Consequently, each module in ARHS-CNN jointly contributes to boosting the performance ARHS pansharpening.

In addition, we also implement further experiments on the contributions of different parts of the composite loss function, whose results are displayed in Fig. 19. Herein, we remove the multistage constraint L_{stage}, the spatial constraint L_{spatial} and the down-sampling reconstruction constraint $L^{\downarrow}_{\text{recon}}$ respectively. Equivalently, we set α_1, α_2, and γ to zero, respectively, which act as the trade-off parameters

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of (7) and (8). It is easy to find that the scores of different metrics obviously drops when the multistage constraint is removed. This phenomenon can be explained by the fact that the multistage constraint directly guides the optimization of parameters from the bottom layers, which mitigates the difficulty of training deep neural networks. Besides, the removal of other constraints also sacrifices the quality of ARHS images to some extent. Therefore, each constraint proposed in our composite loss function is advantageous to the final performance.

V. Conclusion

In this work, we have defined a new type of pansharpening task, i.e., ARHS pansharpening, and then designed a novel CNN method, i.e., ARHS-CNN, to fulfill this task. ARHS pansharpening is significantly different from the traditional HS pansharpening. It aims at pansharpening a LRHS image to any user-customized spatial resolution, which is usually not predefined. As a result, it faces two new challenges. One is how to yield HS images with any resolutions under a single CNN model. The other is how to keep spectral fidelity at those resolutions, which may significantly deviate from the PAN resolution. To address these challenges, our ARHS-CNN comprises two major subnetworks: 1) a multilevel enhancement subnetwork and 2) a rescaling subnetwork. The former subnetwork incorporates the input LRHS and PAN images to yield intermediate spectral–spatial features with the same resolution as the PAN image, fully and directly exploring the spatial structures of the PAN image and thus relieving the burden of computing intermediate features (avoiding spectral interference). In the latter subnetwork, an offset-kernel mapping mechanism is introduced to generate spatially dynamic kernels and, hence, achieving the goal of spatial resolution adjustment. The offset-kernel mapping mechanism (and the related spatially dynamic convolution) guarantees the obtention of data with arbitrary resolutions under a single trained model. After the dynamic convolution, the spectral patterns distorted therein are then recovered in a spectral augmentation operation. Additionally, a composite loss—able to strengthen the ARHS reconstruction—is introduced. As shown in experiments, our ARHS-CNN is able to pansharpen HS images to any spatial resolutions under a single trained CNN model while keeping high spectral fidelity at those resolutions, thus achieving excellent performance in the task of ARHS pansharpening.

References

[1] G. Shaw and D. Manolakis, "Signal processing for hyperspectral image exploitation," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 12–16, Jan. 2002.

[2] L. Loncan *et al.*, "Hyperspectral pansharpening: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[3] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2018.

[4] X. Xu, J. Li, S. Li, and A. Plaza, "Generalized morphological component analysis for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2817–2832, Apr. 2020.

[5] N. M. Nasrabadi, "Hyperspectral target detection : An overview of current and future challenges," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 34–44, Jan. 2014.

[6] D. Marinelli, F. Bovolo, and L. Bruzzone, "A novel change detection method for multitemporal hyperspectral images based on binary hyperspectral change vectors," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4913–4928, Jul. 2019.

[7] N. Yokoya, C. Grohnfeldt, and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 2, pp. 29–56, Jun. 2017.

[8] V. P. Shah, N. H. Younan, and R. L. King, "An efficient pan-sharpening method via a combined adaptive PCA approach and contourlets," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1323–1335, May 2008.

[9] T.-M. Tu, P. S. Huang, C.-L. Hung, and C.-P. Chang, "A fast intensity-hue-saturation fusion technique with spectral adjustment for IKONOS imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 1, no. 4, pp. 309–312, Oct. 2004.

[10] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS + pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Oct. 2007.

[11] G. P. Nason and B. W. Silverman, "The stationary wavelet transform and some statistical applications," in *Wavelets and Statistics*, vol. 103, A. Antoniadis and G. Oppenheim, Eds. New York, NY, USA: Springer-Verlag, 1995, pp. 281–299.

[12] M. Shensa, "The discrete wavelet transform: Wedding the a trous and Mallat algorithms," *IEEE Trans. Signal Process.*, vol. 40, no. 10, pp. 2464–2482, Oct. 1992.

[13] J. G. Liu, "Smoothing filter-based intensity modulation: A spectral preserve image fusion technique for improving spatial details," *Int. J. Remote Sens.*, vol. 21, no. 18, pp. 3461–3472, Dec. 2000.

[14] B. Aiazzi, L. Alparone, S. Baronti, A. Garzelli, and M. Selva, "MTF-tailored multiscale fusion of high-resolution MS and PAN imagery," *Photogramm. Eng. Remote Sens.*, vol. 72, no. 5, pp. 591–596, May 2015.

[15] J. Lee and C. Lee, "Fast and efficient panchromatic sharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 1, pp. 155–163, Jan. 2010.

[16] L. Wenzhi *et al.*, "Processing of multiresolution thermal hyperspectral and digital color data: Outcome of the 2014 IEEE GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2984–2996, Jun. 2015.

[17] A. Garzelli, "A review of image fusion algorithms based on the super-resolution paradigm," *Remote Sens.*, vol. 8, no. 10, pp. 797–816, Sep. 2016.

[18] Q. Wei, N. Dobigeon, and J. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

[19] Q. Wei, N. Dobigeon, and J. Y. Tourneret, "Bayesian fusion of multi-band images," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 6, pp. 1117–1127, Sep. 2015.

[20] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[21] S. Li and B. Yang, "A new pan-sharpening method using a compressed sensing technique," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 2, pp. 738–746, Feb. 2011.

[22] S. Li, H. Yin, and L. Fang, "Remote sensing image fusion via sparse representations over learned dictionaries," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 9, pp. 4779–4789, Sep. 2013.

[23] X. X. Zhu, C. Grohnfeld, and R. Bamler, "Exploiting joint sparsity for pansharpening: The J-SparseFI algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2664–2681, May 2016.

[24] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[25] S. Takeyama, S. Ono, and I. Kumazawa, "Robust and effective hyperspectral pansharpening using spatio–spectral total variation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2018, pp. 1603–1607.

[26] F. Ma, F. Yang, and Y. Wang, "Low-rank tensor decomposition with smooth and sparse regularization for hyperspectral and multispectral data fusion," *IEEE Access*, vol. 8, pp. 129842–129856, 2020.

[27] X. Han, J. Yu, J.-H. Xue, and W. Sun, "Hyperspectral and multispectral image fusion using optimized twin dictionaries," *IEEE Trans. Image Process.*, vol. 29, pp. 4709–4720, 2020.

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[28] G. Masi, D. Cozzolino, L. Verdoliva, and G. Scarpa, "Pansharpening by convolutional neural networks," *Remote Sens.*, vol. 8, no. 7, p. 594, Jul. 2016.

[29] Y. Wei, Q. Yuan, H. Shen, and L. Zhang, "Boosting the accuracy of multispectral image pansharpening by learning a deep residual network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1795–1799, Oct. 2017.

[30] Q. Yuan, Y. Wei, X. Meng, H. Shen, and L. Zhang, "A multiscale and multidepth convolutional neural network for remote sensing imagery pan-sharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 978–989, Mar. 2018.

[31] G. Scarpa, S. Vitale, and D. Cozzolino, "Target-adaptive CNN-based pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5443–5457, Sep. 2018.

[32] L. He *et al.*, "Pansharpening via detail injection based convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1188–1204, Apr. 2019.

[33] L. He, J. Zhu, J. Li, A. Plaza, J. Chanussot, and B. Li, "HyperPNN: Hyperspectral pansharpening via spectrally predictive convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 3092–3100, Aug. 2019.

[34] K. Li, W. Xie, Q. Du, and Y. Li, "DDLPS: Detail-based deep Laplacian pansharpening for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8011–8025, Oct. 2019.

[35] G. He, J. Zhong, J. Lei, Y. Li, and W. Xie, "Hyperspectral pansharpening based on spectral constrained adversarial autoencoder," *Remote Sens.*, vol. 11, no. 22, p. 2691, Nov. 2019.

[36] W. Xie, J. Lei, Y. Cui, Y. Li, and Q. Du, "Hyperspectral pansharpening with deep priors," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1529–1543, May 2020.

[37] G. Vivone *et al.*, "A new benchmark based on recent advances in multispectral pansharpening: Revisiting pansharpening with classical and emerging pansharpening methods," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 53–81, Mar. 2021.

[38] L. He, J. Zhu, J. Li, D. Meng, J. Chanussot, and A. Plaza, "Spectral-fidelity convolutional neural networks for hyperspectral pansharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5898–5914, Sep. 2020.

[39] F. Laporterie-Déjean, H. de Boissezon, G. Flouzat, and M.-J. Lefèvre-Fonollosa, "Thematic and statistical evaluations of five panchromatic/multispectral fusion methods on simulated PLEIADES-HR images," *Inf. Fusion*, vol. 6, no. 3, pp. 193–212, Sep. 2005.

[40] P. Wang, L. Wang, Y. Wu, and H. Leung, "Utilizing pansharpening technique to produce sub-pixel resolution thematic map from coarse remote sensing image," *Remote Sens.*, vol. 10, no. 6, p. 884, Jun. 2018.

[41] P. Sirguey, R. Mathieu, Y. Arnaud, M. M. Khan, and J. Chanussot, "Improving MODIS spatial resolution for snow mapping using wavelet fusion and ARSIS concept," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 1, pp. 78–82, Jan. 2008.

[42] F. Bovolo, L. Bruzzone, L. Capobianco, A. Garzelli, S. Marchesi, and F. Nencini, "Analysis of the effects of pansharpening in change detection on VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 1, pp. 53–57, Jan. 2009.

[43] P. Du, S. Liu, J. Xia, and Y. Zhao, "Information fusion techniques for change detection from multi-temporal remote sensing images," *Inf. Fusion*, vol. 14, no. 1, pp. 19–27, Jan. 2013.

[44] Y. Qu, H. Qi, B. Ayhan, C. Kwan, and R. Kidd, "DOES multispectral/hyperspectral pansharpening improve the performance of anomaly detection?" in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 6130–6133.

[45] A. Mohammadzadeh, A. Tavakoli, and M. J. Valadan Zoej, "Road extraction based on fuzzy logic and mathematical morphology from pan-sharpened IKONOS images," *Photogramm. Rec.*, vol. 21, no. 113, pp. 44–60, Mar. 2006.

[46] Y. Xu, S. E. Smith, S. Grunwald, A. Abd-Elrahman, and S. P. Wani, "Incorporation of satellite remote sensing pan-sharpened imagery into digital soil prediction and mapping models to characterize soil property variability in small agricultural fields," *ISPRS J. Photogramm. Remote Sens.*, vol. 123, pp. 1–19, Jan. 2017.

[47] A. Jenerowicz and M. Woroszkiewicz, "The pan-sharpening of satellite and UAV imagery for agricultural applications," *Proc. SPIE*, vol. 9998, Oct. 2016, Art. no. 99981S.

[48] E. Ibarrola-Ulzurrun, C. Gonzalo-Martín, and J. Marcello, "Influence of pansharpening in obtaining accurate vegetation maps," *Can. J. Remote Sens.*, vol. 43, no. 6, pp. 528–544, 2017.

[49] X. Meng, H. Shen, H. Li, L. Zhang, and R. Fu, "Review of the pansharpening methods for remote sensing images based on the idea of meta-analysis: Practical discussion and challenges," *Inf. Fusion*, vol. 46, pp. 102–113, Jun. 2018.

[50] J. B. Adams and A. R. Gillespie, *Remote Sensing of Landscapes With Spectral Images: A Physical Modeling Approach*. Cambridge, U.K.: Cambridge Univ. Press, 2006.

[51] D. G. Manolakis, R. B. Lockwood, and T. W. Cooley, *Hyperspectral Imaging Remote Sensing: Physics, Sensors, and Algorithms*. Cambridge, U.K.: Cambridge Univ. Press, 2016.

[52] Y. Xie, Z. Sha, and M. Yu, "Remote sensing imagery in vegetation mapping: A review," *J. Plant Ecol.*, vol. 1, no. 1, pp. 9–23, Mar. 2008.

[53] D. Haboudane, J. R. Miller, E. Pattey, P. J. Zarco-Tejada, and I. B. Strachan, "Hyperspectral vegetation indices and novel algorithms for predicting green LAI of crop canopies: Modeling and validation in the context of precision agriculture," *Remote Sens. Environ.*, vol. 90, no. 3, pp. 337–352, 2004.

[54] C. Justice, A. Belward, J. Morisette, P. Lewis, J. Privette, and F. Baret, "Developments in the 'validation' of satellite sensor products for the study of the land surface," *Int. J. Remote Sens.*, vol. 21, no. 17, pp. 3383–3390, Jan. 2000.

[55] Y. Li, W. Xie, and H. Li, "Hyperspectral image reconstruction by deep convolutional neural network for classification," *Pattern Recognit.*, vol. 63, pp. 371–383, Mar. 2017.

[56] N. C. Rakotonirina and A. Rasoanaivo, "ESRGAN+ : Further improving enhanced super-resolution generative adversarial network," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 63–70.

[57] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2015.

[58] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2472–2481.

[59] J. Liu, W. Zhang, Y. Tang, J. Tang, and G. Wu, "Residual feature aggregation network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2356–2365.

[60] H. Zhao, O. Gallo, I. Frosio, and J. Kautz, "Loss functions for image restoration with neural networks," *IEEE Trans. Comput. Imag.*, vol. 3, no. 1, pp. 47–57, Mar. 2017.

[61] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1132–1140.

[62] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogramm. Eng. Remote Sens.*, vol. 63, no. 6, pp. 691–699, 1997.

[63] J. Marcello, E. Ibarrola-Ulzurrun, C. Gonzalo-Martín, J. Chanussot, and G. Vivone, "Assessment of hyperspectral sharpening methods for the monitoring of natural areas using multiplatform remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 2808–2822, Oct. 2019.

[64] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[65] R. H. Yuhas, A. F. Goetz, and J. W. Boardman, "Discrimination among semi-arid landscape endmembers using the spectral angle mapper (SAM) algorithm," in *Proc. Summaries 3rd Annu. JPL Airborne Geosci. Workshop*, vol. 1. 1992, pp. 147–149.

[66] L. Wald, *Data Fusion: Definitions Architectures: Fusion Images Different Spatial Resolutions*. Paris, France: Les Presses de 1' École des Mines, 2002.

[67] A. Garzelli and F. Nencini, "Hypercomplex quality assessment of multi/hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 662–665, Oct. 2009.

[68] L. Alparone, B. Aiazzi, S. Baronti, A. Garzelli, F. Nencini, and M. Selva, "Multispectral and panchromatic data fusion assessment without reference," *Photogramm. Eng. Remote Sens.*, vol. 74, no. 2, pp. 193–200, Feb. 2008.

[69] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent.*, 2014, pp. 1–15.

[70] F. Palsson, J. R. Sveinsson, M. O. Ulfarsson, and J. A. Benediktsson, "Quantitative quality evaluation of pansharpened imagery: Consistency versus synthesis," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1247–1259, Mar. 2016.

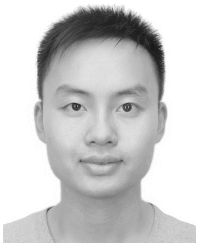
[71] G. Vivone, M. D. Mura, A. Garzelli, and F. Pacifici, "A benchmarking protocol for pansharpening: Dataset, preprocessing, and quality assessment," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6102–6118, Jun. 2021.

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